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An Extension of the Generalized Covariance Measure for
Conditional Independence Testing

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List of abbreviations

AD	Alzheimer’s Disease
ADNI	Alzheimer’s Disease Neuroimaging Initiative
ANOVA	Analysis of Variance
AIC	Akaike Information Criterion
ALASSO	Absolute Shrinkage and Selection Operator
BIC	Bayesian Information Criterion
BLUP	Best Unbiased Linear Predictor
CCIT	Classifier-Based Conditional Independence Test
CDIT	Conditional Distance Independence Test
CDR-SB	Clinical Dementia Rating Scale Sum of Boxes
CPT	Conditional Permutation Test
CRT	Conditional Randomization Tests
CI	Conditional Independence
CIT	Conditional Independence Test
CLT	Central Limit Theorem
CMiknn	Conditional Mutual Information based on Nearest-Neighbor Estimation
DGCIT	Double Generative Adversarial Networks for Conditional Independence Testing
EGCM	Extension of the Generalised Covariance Measure
FCIT	Fast Conditional Independence Test
FDR	False Discovery Rate
GAM	Generalized Additive Model
GAMM	Generalized Additive Mixed Model
GAN	Generative Adversarial Network
GCIT	Generative Adversarial Network for Conditional Independence Testing
GCM	Generalized Covariance Measure
GCV	Generalized Cross-Validation
GEE	Generalized Estimating Equation
GHCM	Generalized Hilbertian Covariance Measure
GLM	Generalized Linear Model
GLMM	Generalized Linear Mixed Model
i.i.d.	independent and identically distributed
KNN	K-Nearest-Neighbor
KCIT	Kernel-based Conditional Independence Test
LASSO	Least Absolute Shrinkage and Selection Operator

Contents

LMEM	Linear Mixed Effects Model
MEM	Mixed Effects Model
ML	Maximum Likelihood
MMCIT	Minimax Optimal Conditional Independence Testing
MMD	Maximum Mean Discrepancy
MMSE	Mini-Mental State Examination
MoCA	Montreal Cognitive Assessment
MRI	Magnetic Resonance Imaging
MSPE	Mean Squared Prediction Error
non-i.i.d.	non-independently and identically distributed
NMEM	Nonlinear Mixed Effects Model
KCIPT	Permutation-based Kernel Conditional Independence Test
PCIT	Predictive Conditional Independence Test
PCM	Projected Covariance Measure
PET	Positron Emission Tomography
PQL	Penalized Quasi-Likelihood
QR-PCT	Partial Copulas Test based on Quantile Regression
RCIT	Randomized Conditional Independence Test
RCoT	Randomized Conditional Correlation Test
REML	Restricted Maximum Likelihood
RKHS	Reproducing Kernel Hilbert Space
SCAD	Smoothly Clipped Absolute Deviation
wGCM	Weighted Generalised Covariance Measure

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1 Introduction

Understanding the concept of conditional independence (CI), defined as the scenario where two variables are independent given a third variable, is fundamental in various domains of statistical analysis. CI testing plays a pivotal role in graphical models (Campos & Huete 2000; Koller & Friedman n.d.), simplifying the complex probabilistic relationships; in causal inference (Spohn 1994), identifying causal relationships from observational data; and in feature selection (Xing et al. 2001), reducing dimensionality by selecting relevant variables.

Beyond theoretical statistics, CI testing is a critical component, ensuring the robustness of scientific results in genetics, medicine, and economics studies (Barut et al. 2012; Linden & Glas 2010; K. Zhang et al. 2012; Nogales et al. 2000; Ślęzak 2009). For instance, in medical diagnosis, innovative diagnostic techniques such as magnetic resonance imaging (MRI) offer significant advantages over traditional cognitive assessments in early disease detection (Silverberg et al. 2011; Loewenstein et al. 2018; Mortamais et al. 2017; Hobart et al. 2013), particularly in the detection of patients with Alzheimer’s disease (AD) (Popuri et al. 2020; Khatri & Kwon 2022; Leandrou et al. 2018; Olsen et al. 2017). CI testing can potentially confirm the superior efficacy of these innovative diagnostic tools over traditional cognitive tests in clinical applications.

The scope of conditional independence tests (CITs) has seen substantial expansion in recent years. Contemporary nonparametric CITs have diversified into four main categories: metric-based tests, kernel-based tests, simulation-based tests, and regression-based tests. Each category evaluates CI from unique perspectives. Specifically, metric-based tests utilize distance correlations, information metrics, or predictability between conditional and joint distributions to evaluate CI; kernel-based tests examine CI in the Reproducing Kernel Hilbert Space (RKHS); simulation-based tests generate simulations based on the CI assumption for comparison against the empirical distribution; and regression-based tests perform separate regressions on a third variable against each of the other two variables and examine the residuals for signs of independence (C. Li & Fan 2020; Polo, Sun, & Banerjee 2023).

Despite the wide applicability of CI testing, it still faces many challenges. For continuous

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variables X , Y , and Z , constructing a CIT that controls the Type-I error simultaneously at a predetermined significance level α across all distributions under the null hypothesis, while demonstrating greater power than α for any alternative hypothesis, remains a challenging task (Shah & Peters 2020). Even local CIT encounters difficulties (Kim et al. 2022).

Furthermore, predominant nonparametric CITs presuppose that data are independent and identically distributed (i.i.d.), a condition seldom encountered in real-world scenarios. This may result in bias when handling non-i.i.d. data (C. Li & Fan 2020; Flaxman et al. 2015), as exemplified by the repeated measurements structure of the Alzheimer’s Disease Neuroimaging Initiative (ADNI) dataset. Although several studies have proposed nonparametric CITs designed for non-i.i.d. data, aiming to improve CI testing accuracy by accounting for the dependence structure, However, most of these studies focus on time series data, often overlook the special case of repeated measurements (Bouezmarni et al. 2012; X. Wang & Hong 2018; Flaxman et al. 2015). Analyzing repeated measurements is crucial for understanding temporal dynamics, enhancing data completeness, and improving the quality and reliability of statistical inference (Diggle 1988; Muhammad 2023; Fixsen et al. 2023). Repeated measurements data in the ADNI dataset involve multiple measurements obtained from the same subjects over time (ADNI Team 2023). This intuitively creates a correlation within measurements from the same subject, which diverges from the i.i.d. assumption. As such, conventional CITs, which are primarily designed for i.i.d. data, may fail to control the Type I error rate and potentially lack power when analyzing dependent data (C. Li & Fan 2020; Flaxman et al. 2015). To bridge this gap, the goal of this thesis is to explore the possibility of extending CI testing to better suit the analysis of repeated measurements and to provide a framework that accounts for the unique characteristics of repeated measurements, which could enhance CI testing performance in dependent data.

For our research aim, we reviewed the current CITs and chose the framework of the Generalized Covariance Measure (GCM) (Shah & Peters 2020) due to its regression-based flexibility. We then analyzed models that are appropriate for repeated measurements and chose the adaptable Generalized Additive Mixed Model (GAMM) (Groll & Tutz 2011, 2012). By extending the GCM framework to include GAMM, we introduced an Extension of the Generalized Covariance Measure (EGCM), and aimed to address limitations of existing CITs in analyzing repeated measurements. Through simulations and real-world applications using the ADNI dataset, this thesis demonstrated the advantages of EGCM in non-i.i.d. data. The results indicate the potential of EGCM to uncover correlations within repeated measurements, therefore improving the understanding of AD through the relationship analysis of structural MRI markers and cognitive assessment scores.

1 Introduction

This thesis is structured to explore our research aim comprehensively, starting with an introduction to the problem in Chapter 1 and an extensive literature review on related CITs in Chapter 2. The possible applicability of the GCM for dependent data is discussed, after a detailed discussion of its fundamentals, theoretical basis, and current regression methods in Chapter 3. Chapter 4 focuses on mixed effects models (MEMs), and GAMM is chosen as the ideal candidate to expand the GCM framework. Afterwards, Chapter 5 thoroughly presents the EGCM framework, its test statistic, and the critical role of residual analysis in CI testing. It concludes in Chapter 7 by summarizing the contributions of the EGCM to the development of CI testing in repeated measurements data, and discusses potential future research directions.

By extending the GCM framework to incorporate GAMM, the EGCM can address limitations commonly encountered in existing CITs when analyzing dependent data. This extension is likely to enhance the theoretical robustness of CI testing and potentially demonstrate practical applicability in uncovering complex conditional dependence relationships within the ADNI dataset.

2 Literature Review

Investigating CI testing is fundamental in modern statistical analysis and machine learning, connecting theoretical foundations with real-world applications across a variety of scientific fields (Hoyer et al. 2009; Peters et al. 2014; Strobl et al. 2019; Glymour et al. 2019; Candes et al. 2018; Watson & Wright 2021; Ritov et al. 2017; Polo, Izbicki, et al. 2023). This chapter provides an extensive review of the literature on CITs, building a strong foundation for understanding their current state. It explores the evolution of CITs from traditional parametric methods to the latest nonparametric approaches. Additionally, it includes a critical discussion on existing CITs in the context of dependent data structures. Through this exploration, we offer an overview of CITs' development, acknowledging creative contributions and potential constraints. This sets the stage for a detailed examination of extending a specific CIT for CI testing in non-i.i.d. data.

2.1 Defining Conditional Independence

Before our comprehensive review of CITs, we begin with the foundational concept of CI in statistical analysis. It provides a framework for understanding the relationships among variables within a conditioned environment. It simplifies models by reducing the number of parameters needed to describe the joint distribution of a set of variables (Basu & Pereira 2011), improves understanding of causal relationships between variables (Pearl 2000), and is a cornerstone of Bayesian network learning (Lerner et al. 2013).

Consider a random vector (X, Y, Z) with values in $\mathcal{X} \times \mathcal{Y} \times \mathcal{Z} \subseteq \mathbb{R}^{d_X \times d_Y \times d_Z}$, and let $\mathcal{P}$ denote a specific family of distributions over the measurable space $(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z}, \mathcal{B})$, where $\mathcal{B} = \mathcal{B}(\mathcal{X} \times \mathcal{Y} \times \mathcal{Z})$ represents the Borel σ -algebra. Assuming $(X, Y, Z) \sim P$ with P belonging to $\mathcal{P}$, we define $\mathcal{P}_0$ as the subset of $\mathcal{P}$ for which X is independent of Y given Z . The CI testing is formulated as:

$$H_0 : P \in \mathcal{P}_0$$

$$H_1 : P \notin \mathcal{P}_0$$

In simpler terms, this is expressed as: $H_0 : X \perp\!\!\!\perp Y \mid Z$ and $H_1 : X \not\perp\!\!\!\perp Y \mid Z$. $H_0 : X \perp\!\!\!\perp$

$Y \mid Z$ indicates the hypothesis of CI that X and Y are independent given $Z = z$, for any value of z , whereas $H_1 : X \not\perp Y \mid Z$ indicates the alternative hypothesis where such independence does not exist (Dawid 1979; Shah & Peters 2020; Polo, Sun, & Banerjee 2023; Scheidegger et al. 2022). CI occurs when the knowledge of one variable does not influence the relationship between the other two variables. Intuitively, this means that knowledge of the confounding variables Z makes two random variables X and Y unrelated, removing any predictive ability X might have had on Y , and vice versa.

CI is typically defined within a probabilistic framework. Let X , Y and Z be continuous random variables within the probability space $(\Omega, \mathcal{F}, P)$ within which the random variables X , Y , and Z are defined. Ω is the sample space; containing all possible outcomes; $\mathcal{F}$ is a σ -algebra of events; and P is a probability measure that assigns a probability to each event in $\mathcal{F}$. We denote by $p_{X,Y|Z}(x, y|z)$, $p_{X|Z}(x|z)$ and $p_{Y|Z}(y|z)$, respectively, the conditional joint density of (X, Y) given $Z = z$, the marginal density of X given $Z = z$, and the marginal density of Y given $Z = z$. CI for continuous variables X and Y given Z is defined as:

$$p_{X,Y|Z}(x, y|z) = p_{X|Z}(x|z) \cdot p_{Y|Z}(y|z).$$

The first equation states that the conditional joint density of X and Y given Z is equal to the product of the marginal densities of X given Z and Y given Z (Dawid 1979). This equation forms the foundation of CI. This concept is equivalently represented in the following forms:

$$\begin{aligned} p_{X|Y,Z}(x|y, z) &= p_{X|Z}(x|z); \\ p_{Y|X,Z}(y|x, z) &= p_{Y|Z}(y|z); \\ p_{XY|Z}(x, y|z) &= p_{X|Z}(x|z) \cdot p_{Y|Z}(y|z). \end{aligned}$$

The equivalent expressions provide alternative formulations that further elucidate the concept of CI. These expressions state that the conditional density of one variable given the others is equivalent to its conditional density given the conditioning set alone (Scheidegger et al. 2022; C. Li & Fan 2020; Polo, Sun, & Banerjee 2023).

The alternative characterization of CI transforms the CI into independence within certain L^2 spaces (Daudin 1980) and circumvents the potential complexities involved in directly estimating densities, offering a more abstract yet powerful framework for understanding CI (C. Li & Fan 2020). In probability theory, an $L^2(\Omega, \mathcal{F}, P)$ space comprises random

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variables with finite second moments. Specifically, constrained L^2 spaces are defined as follows:

$$\begin{aligned}\mathcal{F}_{XZ} &\equiv \{f \in L_{XZ}^2 : \mathbb{E}[f(Z) | X] = 0\}, \\ \mathcal{F}_{YZ} &\equiv \{g \in L_{YZ}^2 : \mathbb{E}[g(Z) | Y] = 0\}, \\ \mathcal{F}'_{XZ} &\equiv \{f' : f' = f(X) - \mathbb{E}[f(Z) | Y], f' \in L_X^2\}, \\ \mathcal{F}'_{YZ} &\equiv \{g' : g' = g(Y) - \mathbb{E}[g(Z) | Y], g' \in L_Y^2\}.\end{aligned}$$

where L_{XZ}^2 and L_{YZ}^2 represent the spaces of square-integrable functions of random variables X with Z , and Y with Z , respectively. Functions in these spaces are mappings involving these random variables that satisfy the square-integrability condition under the probability measure. $\mathcal{F}_{XZ}$ and $\mathcal{F}_{YZ}$ are subspaces within these L^2 spaces comprising regression functions that have an expected value of zero when conditioned on one variable. This condition implies that, after conditioning on X for $\mathcal{F}_{XZ}$ or Y for $\mathcal{F}_{YZ}$, the function f or g does not systematically vary with Z , illustrating a form of independence in terms of expected values. $\mathcal{F}'_{XZ}$ or $\mathcal{F}'_{YZ}$ further refines this concept by defining a new regression function f' or g' from f or g in L_{XZ}^2 or L_{YZ}^2 by subtracting its expected value conditioned on X or Y , focusing solely on the uncorrelated nature of the residual function. Assuming the residual functions $\tilde{f}'$ and $\tilde{g}'$ result from the regression of $\tilde{f}$ and $\tilde{g}$, the following conditions (i) to (v) are equivalent:

- (i) $X \perp\!\!\!\perp Y | Z$;
- (ii) $\mathbb{E}(\tilde{f}\tilde{g}) = 0, \forall \tilde{f} \in \mathcal{F}_{XZ} \text{ and } \tilde{g} \in \mathcal{F}_{YZ}$;
- (iii) $\mathbb{E}(\tilde{f}g) = 0, \forall \tilde{f} \in \mathcal{F}_{XZ} \text{ and } g \in L_{YZ}^2$;
- (iv) $\mathbb{E}(\tilde{f}\tilde{g}') = 0, \forall \tilde{f} \in \mathcal{F}_{XZ} \text{ and } \tilde{g}' \in \mathcal{F}'_{YZ}$;
- (v) $\mathbb{E}(\tilde{f}g') = 0, \forall \tilde{f} \in \mathcal{F}_{XZ} \text{ and } g' \in L_Y^2$.

Conditions (ii) to (v) translate the CI concept into zero expected conditional covariance between any regression functions and its residuals functions within the defined L^2 spaces (Daudin 1980). This characterization of CI can be understood as an extension of the partial correlation principle. In the classical statistical framework, partial correlation $\rho_{XY|Z}$ measures the correlation between two variables X and Y , while controlling for a third variable Z . When $\rho_{XY|Z} = 0$, it indicates that any correlation between X and Y can be entirely explained by their respective relationships with Z , hence implying CI. In

cases where X , Y , and Z follow a joint Gaussian distribution, condition (ii) equates to a zero partial correlation coefficient (K. Zhang et al. 2012; C. Li & Fan 2020). Nevertheless, the concept of CI, within the context of expected covariance between regression functions and their residuals in L^2 spaces, captures the essence of CI more broadly than partial correlation by encompassing a wider array of functional relationships among the variables. The partial cross-covariance operator within characteristic kernel spaces potentially offers a more compact framework than encompassing all functions in L^2 (Fukumizu et al. 2008).

In this section, we explore CI as a key concept in statistics with wide-ranging applications. It is impractical to check every probability density of X and Y given Z in a continuous, infinite sample space. Consequently, more flexible approaches for determining CI have been developed (C. Li & Fan 2020). Without directly estimating densities, CI can be inferred by examining the covariance of expected regression functions between variables. If the covariance of these expectation functions is zero, this suggests that X and Y do not have additional correlation beyond what is explained by Z , thus supporting their CI. Another approach involves checking the covariance of residual functions. If the expected covariance between the residuals of X regressed on Z and Y regressed on Z is zero, X and Y can be considered conditionally independent given Z . In scenarios where the variables follow a joint Gaussian distribution, the partial correlation coefficient provides a means to assess CI. A zero partial correlation coefficient between X and Y given Z indicates no linear association between X and Y once Z is controlled for. With these approaches to characterizing CI, the development of CITs aims to more accurately differentiate between CI and non-CI hypotheses. As the fields of statistics and machine learning evolve, researchers have proposed various CIT methods to accommodate different data structures and assumptions. In the next section, a brief overview of the current state of CITs will be provided.

2.2 Evolving Landscape of Conditional Independence Tests

The common goal of CITs is to robustly and effectively utilize various CI characterizations to test for CI between variables in data. In the following section, we will delve into the primary categories of CITs and their advancements in practical applications, including the strengths and limitations of these tests, their appropriate contexts, and the potential challenges they face in modern data analysis.

CI testing is crucial in various fields, including feature selection (Barut et al. 2012; Linden & Glas 2010; Candes et al. 2018; Watson & Wright 2021; Xing et al. 2001), causal inference

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(Hoyer et al. 2009; Peters et al. 2014; Strobl et al. 2019; Glymour et al. 2019; K. Zhang et al. 2011), Bayesian networks (Nogales et al. 2000; Ślęzak 2009) and machine learning (Ritov et al. 2017; Polo, Izbicki, et al. 2023), providing insights into the relationships between variables. Despite its importance, CI testing encounters several challenges, particularly when dealing with different types of conditional variables Z .

For a discrete conditional variable Z , traditional tests such as Pearson’s χ^2 and likelihood ratio tests have their limitations (Edwards 2000; Pearson 1992). These tests struggle with complexities inherent to CI testing, especially when Z has many levels or when data is sparse. An alternative approach (Margaritis 2005) transforms CI testing into unconditional formats through discretization for discrete Z , but this requires large sample sizes to maintain accuracy (C. Li & Fan 2020).

When the conditional variable Z is continuous, CI testing diverges into two main approaches: parametric and nonparametric methods (Bergsma 2004). Parametric methods are known for their computational efficiency, these methods rely on specific assumptions about the data’s distribution (Fisher 1915; Pearson 1992). Their effectiveness diminishes when assumptions about linearity, additivity, and normality do not hold, revealing a critical vulnerability in traditional parametric approaches (Daudin 1980; Peters et al. 2014). Nonparametric methods offer greater flexibility as they do not presuppose specific functional relationships or distribution forms for the variables involved. However, this adaptability increases computational demands, particularly with high-dimensional data or large sample sizes. Managing this without imposing assumptions remains a persistent challenge, leading to significant computational overhead (Bergsma 2004; K. Zhang et al. 2012). The integration of CI testing in multivariate datasets or with supervised learning algorithms introduces additional complexity, potentially leading to model misspecification and failure in controlling Type I errors (Polo, Sun, & Banerjee 2023).

The challenges associated with nonparametric CITs emphasize some of the inherent limitations in CI testing methodologies. A fundamental constraint is that it is impossible to design a CIT that simultaneously achieves both objectives: controlling the Type-I error at a predetermined significance level α for all distributions under the null hypothesis (i.e., when CI holds) and exhibiting greater power than α against any alternative hypothesis for continuous distributions of variables X , Y , and Z . In practice, this translates to a trade-off between ensuring that the CIT does not falsely detect dependencies (minimizing Type-I errors) and the ability of the test to correctly identify true dependencies when they exist (test power) (Shah & Peters 2020). Local permutation tests for CI within continuous environments are explored more deeply in this discussion, which involve reshuffling data

to create a distribution of test statistics under the null hypothesis. Additional constraints are necessary to for managing Type-I error rates while ensuring that the local CITs retain sufficient power to detect true conditional dependencies (Kim et al. 2022).

Given the limitations and complexities of nonparametric CITs, it is imperative to carefully consider the choice of CITs, recognizing that no single test may be universally optimal (Berrett et al. 2020). The selection of a CIT should take into account the unique characteristics of the data and the specific research question being addressed. Ongoing methodological innovations have sought to overcome these limitations by developing nonparametric CITs that better balance the trade-offs between Type-I error control and test power, or by adapting existing tests to perform more effectively under certain conditions.

The landscape of nonparametric CITs has indeed broadened significantly in recent years. This expansion has led to a rich variety of methods, each developed to address the challenges of CI testing from unique perspectives (Marx & Vreeken 2019; Shah & Peters 2020; C. Li & Fan 2020; Neykov et al. 2021; Watson & Wright 2021; Kim et al. 2022; Shi et al. 2021; Scetbon et al. 2021; Tansey et al. 2022; H. Zhang et al. 2022; Ai et al. 2022). The categorization of these tests into distinct groups highlights the diversity of approaches and the evolving nature of the field. Broadly, the existing tests can be categorized into four main groups: metric-based tests, kernel-based tests, simulation-based tests, and regression-based tests. Beyond these primary categories, there are other innovative approaches to CI testing, including graphical models, copula measures, or novel statistical techniques designed to address specific challenges in CI testing (Bergsma 2004; Berrett et al. 2020).

In light of the diversity of CITs, we provide a structured summary in Table 2.1 introducing key recent developments in CITs across four main categories. Subsequently, we delve into a detailed discussion of the key methods within each category, outlining their foundational principles and potential applicability. This thorough exploration seeks to identify methods that align with our research objectives.

- **Metric-based tests** utilize various metrics, including distance correlations, information metrics, predictability, and similarity between conditional distributions, and the actual joint distribution to evaluate CI (Su & White 2007, 2014; Chalupka et al. 2018; Burkart & Király 2018; X. Wang et al. 2015; Pan et al. 2017; X. Wang et al. 2018; Runge 2018; Kubkowski et al. 2021; H. Zhang et al. 2022).

Conditional Distance Independence Test (CDIT) (X. Wang et al. 2015) stands out for its distance correlation, offering a flexible and effective approach suitable for

Table 2.1: Overview of nonparametric CITs categorizes methods into four main types: metric-based, kernel-based, simulation-based, and regression-based. Each type is evaluated for core concepts, applicability, and limitations, aiding in the selection of suitable tests.

Category	Method	Main Advancement	Evaluation
Metric-based Tests	CDIT	Distance correlation.	Multivariate analysis.
	CMknn	KNN search & mutual information.	High-dimensional data.
	PCIT	Predictability with expected losses.	Large-scale datasets.
Kernel-based Tests	FCIT	Mean squared error for prediction accuracy.	Large sample datasets.
	KCIT	Partial covariance within RKHS.	Computationally intensive.
	RCoT, RCIT	Fourier features for scalability.	Improves computation.
	KCIPT	Permutation & kernel-based.	Careful Kernel choice.
Simulation-based Tests	CRTs	Data simulation.	Require accurate approximation
	CPTs	Permutes non-uniform.	High-dimensional data.
Regression-based Tests	GCM	Covariance of residuals.	Fundamental measure.
	wGCM	Sample splitting & weighting.	Enhanced sensitivity.
	PCM	Data splitting & projection function.	Improves test accuracy.
	GHCM	Functional random variables.	Functional data analysis.
	QR-PCT	Quantile regression & partial copula.	For heterogeneity.
	DGCIT	Integrates GAN.	Large-scale studies.
	MMCIT	From a minimax perspective.	Advances in GCM theory.

multivariate data analysis. This method, leveraging characteristic functions, enables CI exploration across various dimensions.

Conditional Mutual Information based on Nearest-Neighbor Estimation (CMIknn) (Runge 2018) offers a practical solution for quantifying unique information exchange between variables. It employs a k-nearest-neighbor (KNN) search to mitigate the influence of conditioning variables. While this method excels in handling high-dimensional data, it requires robust search algorithms and careful hyperparameter tuning (Jaderberg et al. 2017).

Predictive Conditional Independence Test (PCIT) (Burkart & Király 2018) and Fast Conditional Independence Test (FCIT) (Chalupka et al. 2018) focus on prediction accuracy to evaluate CI. The comparative analysis of expected losses in PCIT and mean squared error in FCIT highlight the predictive performance’s flexibility in evaluating CI across large datasets.

Selecting an appropriate metric-based test and tuning hyperparameters are essential for effectively employing these methodologies. Each test’s unique operational mechanism and underlying assumptions require careful consideration of the specific research context. For example, CMIknn’s implementation may depend on the dataset’s dimensionality and the efficiency of available KNN search algorithms (Runge 2018; Jaderberg et al. 2017).

- **Kernel-based tests** compare embeddings of joint and marginal distributions in the RKHS. These tests are particularly adept at capturing nonlinear dependencies and offer a careful evaluation of CI (Fukumizu et al. 2008; K. Zhang et al. 2012; Doran et al. 2014; Strobl et al. 2019; Scetbon et al. 2021).

Kernel-based Conditional Independence Test (KCIT) (K. Zhang et al. 2012) leverages Maximum Mean Discrepancy (MMD) to quantify the average association between variables X and Y , conditioned on Z (Gretton et al. 2012). This approach allows for detailed CI testing by measuring differences in embeddings of joint and marginal distributions within RKHS. KCIT may face increased Type-I error rates and reduced efficacy in scenarios with small sample sizes or high-dimensional conditioning sets. The complexity of RKHS computations also imply significant computational demands (Strobl et al. 2019; Imaizumi & Fukumizu 2019).

Randomized Conditional Correlation Test (RCoT) and Randomized Conditional Independence Test (RCIT) (Strobl et al. 2019) address some of KCIT’s limitations through the incorporation of random Fourier features to approximate kernel functions. Specifically, shift-invariant kernels can be approximated in lower-dimensional spaces by sampling from their Fourier transforms. These tests aim to improve scalability of kernel-based CI testing, enhancing computational efficiency for large conditioning sets. However, they face challenges with datasets exhibiting complex, nonlinear dependencies, and kernel function approximations might introduce biases (Imaizumi & Fukumizu 2019).

Permutation-based Kernel Conditional Independence Test (KCIPT) (Doran et al. 2014) employs permutation strategies within the kernel-based testing framework. This method aims to simulate data under CI conditions and compare it to observed data, with computational complexity arising from permutation requirements. Like other kernel-based tests, KCIPT’s performance is influenced by kernel choice and conditioning set dimensionality (Gretton et al. 2012).

Selection of a kernel-based test for CI testing should consider the specific data characteristics. For instance, KCIT may be preferred where computational resources are ample, and conditioning set dimensionality is fixed (Shi et al. 2021). RCoT and RCIT may provide practical solutions for large-scale applications with high-dimensional data (Strobl et al. 2019). KCIPT offers a robust permutation-based approach, useful for validating model assumptions or exploring dependency structures’ impacts on the conditional set (Doran et al. 2014). The choice of kernel is critical for the effectiveness of kernel-based tests. Incorrect kernel selections can result in inflated Type I error rates or diminished test power. To ensure asymptotic approximation, kernel-based tests often rely on specific theoretical assumptions about the data and the chosen kernel function. Deviations from these assumptions can significantly impact the validity of test outcomes (Imaizumi & Fukumizu 2019; Shi et al. 2021). Exploring function spaces beyond traditional RKHS, such as those generated by neural networks, could allow new kernel-based tests to achieve greater adaptability in detecting CI (Doran et al. 2014).

- **Simulation-based tests**, including Conditional Randomization Tests (CRTs) (Candes et al. 2018; Bellot & van der Schaar 2019) and Conditional Permutation Tests (CPTs) (Berrett et al. 2020; Tansey et al. 2022), simulate data under the CI null hypothesis by drawing samples from the conditional distributions of X given Z or Y

given Z . These tests aim to approximate conditional distributions either directly or indirectly. By comparing simulated data against actual observations, they evaluate CI. This category encompasses tests relying on resampling techniques, generative models, or other forms of simulation to create reference distributions under the CI null hypothesis. Methods involve generating M sets of simulated data and calculating the proportion of simulations where the test statistic for the simulated data exceeds that of the observed data, serving as the p-value for the test (Candes et al. 2018; Bellot & van der Schaar 2019; Berrett et al. 2020; Shi et al. 2021; Tansey et al. 2022; Liu et al. 2022).

Generative Conditional Independence Test (GCIT) (Bellot & van der Schaar 2019), a CRT, approximates the conditional distributions of X given Z using Generative Adversarial Networks (GANs). By learning a generator function from observed data, GCIT simulates samples that replicate conditional distributions, thus evaluating CI within a generative framework. GCIT encounters challenges in GAN training, including hyperparameter selection (Jaderberg et al. 2017). Its control of Type I error relies on the generative model’s fidelity to true conditional distributions (Shi et al. 2021; Polo, Sun, & Banerjee 2023).

CPT is distinguished by its strategy of non-uniformly permuting X entries, designed to maintain the relationship between X and Z . Unlike CRTs, which estimate the conditional distribution for X given Z to generate new X -values, CPT constructs a non-uniform distribution for permuting observed X -values, allowing it to reflect the true underlying data structure. An efficient Markov Chain Monte Carlo (MCMC) sampler addresses the computational challenges often associated with high-dimensional data and complex dependency structures (Berrett et al. 2020).

Selecting a suitable simulation-based test for CI testing involves a multifaceted decision process that considers the characteristics of the data, computational limitations, analysis objectives, and practical implementation aspects. Tests like CPTs or CRTs may be particularly advantageous for high-dimensional Z , potentially benefiting from dimensionality reduction techniques or efficient sampling methods. However, tests requiring intensive computation, such as those involving complex MCMC sampling or the training of deep generative models, may be less feasible for limited computational resources. These tests strive to balance the need for accurate approximation of conditional distributions against computational efficiency. In scenarios of model

misspecification, there is a risk they might fail to control Type I error (Polo, Sun, & Banerjee 2023).

- **Regression-based tests** employ regression models to approximate the conditional expectations between variables. These tests examine the residuals of such regressions for signs of remaining dependency. This approach can be tailored to various data types, making it flexible and widely applicable. Regression-based tests often incorporate advances in statistical modeling, including nonlinear, quantile regression methods or supervised learning methods, to refine their analysis of CI testing (Peters et al. 2014; Q. Zhang et al. 2017; H. Zhang et al. 2019; Shah & Peters 2020; Dai et al. 2022; Hoyer et al. 2009; Petersen & Hansen 2021; Lundborg et al. 2023; Scheidegger et al. 2022; Neykov et al. 2021).

GCM is foundational, focusing on the sample covariance of residuals from regressions of X on Z and Y on Z to test CI. CI implies that the normalized covariance of residuals should be close to zero. It is suited for cases where a basic yet effective measure of CI is needed (Shah & Peters 2020).

Weighted Generalised Covariance Measure (wGCM) (Scheidegger et al. 2022) improves GCM by incorporating sample splitting and a weighted function, enhancing sensitivity in detecting conditional dependencies in more complex scenarios than GCM.

Projected Covariance Measure (PCM) (Lundborg et al. 2023) further extends GCM by employing a portion of data to estimate projection function through machine learning methods and then calculating the test statistic with the remaining data. The data splitting refines the projection function estimation and improves test accuracy when dealing with high-dimensional data.

Generalised Hilbertian Covariance Measure (GHCM) (Lundborg et al. 2022) adapts GCM for functional random variables, suitable for complex data types like functional graphical models or regression models with functional predictors. GHCM bridges CI testing with functional data analysis.

Partial Copulas Test based on Quantile Regression (QR-PCT) (Petersen & Hansen 2021) uses quantile regression to estimate conditional distribution functions, employing a generalized correlation measure within the partial copula framework. It can handle conditional variance heterogeneity.

2 Literature Review

Double Generative Adversarial Networks for Conditional Independence Testing (DGCIT) (Shi et al. 2021) integrates deep learning with CI testing through a dual-generator GAN architecture to enhance GCM. However, similar to GCIT, it encounters complexities in manual hyperparameter tuning (Jaderberg et al. 2017).

Minimax Optimal Conditional Independence Testing (MMCIT) (Neykov et al. 2021) explores GCM from a minimax perspective, introducing bounds on the minimax critical radius.

Selecting a regression-based test for CI testing involves evaluating the advantages and challenges of each regression method in relation to the research question and data characteristics. These tests encompass two critical steps: regression to estimate conditional means and subsequent analysis to assess the independence of residuals from these regressions. This dual-step approach offers a versatile framework for robust CI testing, accommodating domain-specific insights and cutting-edge regression techniques. The efficacy of regression-based tests hinges on the asymptotic mean of the test statistic under the null hypothesis being zero, and under the alternative hypothesis being nonzero. To date, there are no published robustness results for regression-based CI tests (Berrett et al. 2020). The selection may hinge on the compatibility of each model’s underlying assumptions with the data’s features. For instance, DGCIT may be particularly advantageous in large-scale studies, given its compatibility with deep learning models. However, advanced regression techniques, particularly those involving nonlinear models or high-dimensional data, may pose computational challenges. When computational resources are constrained, simpler approaches like GCM could be more feasible. For intricate dependency structures, PCM or wGCM might enhance Type I error control under CI. Model misspecification can lead to false conclusions, emphasizing the importance of meticulous model choice (Polo, Sun, & Banerjee 2023).

In the scope of previous nonparametric CITs, regression-based tests for CI have significantly evolved, providing a flexible framework capable of adapting to diverse data scenarios through the integration of various regression techniques and machine learning algorithms. This evolution builds on the foundation established by the GCM, with numerous methodologies developed to extend its basic framework to address the complexities of real-world data. While traditionally GCM assumes i.i.d. data, it has shown potential for handling non-i.i.d. data by employing suitable regression methods. Such adaptability is essential, given the widespread presence of non-i.i.d. characteristics in datasets across

fields like biomedical research and social sciences. The following discussion will thoroughly review the literature to examine current CITs in light of the i.i.d. assumption violation due to data dependencies and heterogeneity.

2.3 Challenges in Conditional Independence Testing for Dependent Data

Building on our discussion of the advantages and limitations of nonparametric CITs, this section delves into the diverse methods tailored to address CI testing challenges, particularly under the complexities posed by non-i.i.d. data. These challenges stem from dependencies and variations unaccounted for in previously discussed CITs in Section 2.2, notably in repeated environments where data on the same subjects exhibit serial correlations and disparate observation counts over time. This exploration not only highlights the impacts of non-i.i.d. data on CITs but also reviews specific CIT methodologies designed for dependent data.

In the context of the ADNI dataset, the characteristic of repeated measurements reflects the longitudinal nature of the study, where subjects undergo a series of assessments over time. These measurements provide insight into the progression of AD but introduce dependency structures that challenge traditional CITs due to their non-i.i.d nature. Specifically, Figure 2.1 tracks the Mini-Mental State Examination (MMSE) scores of five randomly selected participants from the ADNI dataset across multiple visits. The bar graph adjacent to the plot summarizes the frequency of score occurrences, illustrating a general trend of declining MMSE scores over time for some participants. Notably, participant 344, as indicated by the labeled data points, consistently exhibits lower MMSE scores compared to other participants. This individual's scores display a downward trend, suggesting potential cognitive impairment progression.

The ADNI dataset's repeated measurements on the same subjects lead to correlations within individuals' data points, described as random effects. These account for the individual-specific variability, recognizing that each subject has a unique intercept (baseline cognitive function level) and possibly unique slopes (rates of cognitive decline). This variability represents structured variation, not merely noise. Systematic influences affecting all individuals similarly in ADNI data could be treated as fixed effects, which are constant within the same subject. These could include demographic factors (gender), or genetic predispositions (genetic information identifies through genome-wide association studies).

Utilizing CITs that assume i.i.d. conditions without adjustments for non-i.i.d. datasets

2 Literature Review

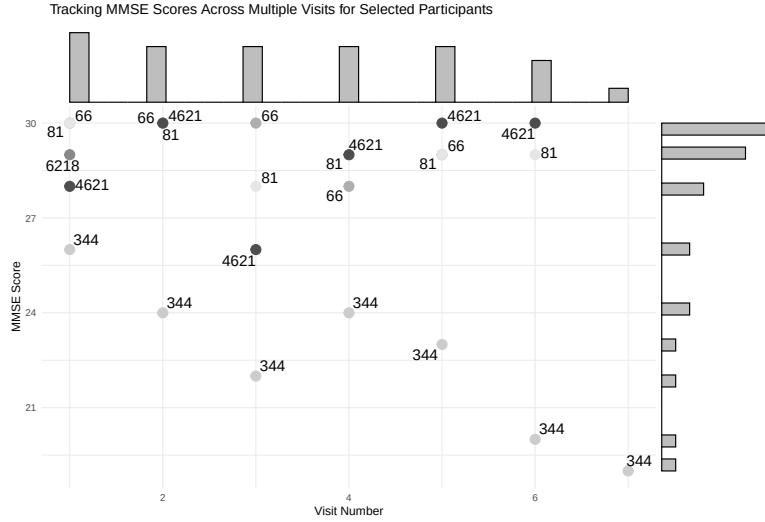


Figure 2.1: The plot illustrates the progression of MMSE scores across repeated assessments for five randomly selected participants from the ADNI dataset. The vertical axis displays MMSE scores, which serve as indicators of cognitive function, with higher scores indicating better cognitive performance. The horizontal axis represents the visit number, arranged chronologically, highlighting the timing of assessments. Data points for each participant emphasize the intra-individual consistency of MMSE scores, a result of the study’s repeated measures design. A discernible trend of declining MMSE scores over time is observed for the participant with ID 344. Additionally, a bar graph adjacent to the plot summarizes the frequency of MMSE scores and visit occurrences, offering further insight into their temporal distributions.

can lead to incorrect test results. For example, when applied to data structures such as repeated measurements in the ADNI dataset, the previous CITs may not account for inherent dependencies and variations. Relying on i.i.d. assumptions in the context of non-linear and non-Gaussian data may introduce systematic biases. These biases, primarily arising because non-i.i.d. settings may disrupt the convergence of test statistics to their expected conditional distributions under the null hypothesis, potentially reduce the reliability and precision of CI testing (Ramsey 2014; C. Li & Fan 2020; Flaxman et al. 2015). These biases necessitate modifications to CITs to ensure the validity of tests in real-world non-i.i.d. data.

The extension of CITs to handle non-i.i.d. data has seen the development of several innovative approaches, each designed to manage specific data dependencies effectively. Next, we examine these methods and their applications, focusing on the types of dependent data they process and the reasons they might not be suitable for repeated measurements data, underscoring the necessity for extending the GCM to bridge this gap.

Smoothed empirical likelihood ratio test (Su & White 2014) applies empirical likelihood to time series data, adeptly uncovering complex causal relationships within time-dependent

structures. The copula-based method model (Bouezmarni et al. 2012) estimates distributions using copula functions, distinguishing itself by capturing dependency structures independently of marginal behaviors, in air pollution research. The characteristic function-based test (X. Wang & Hong 2018) employs characteristic functions for various hypothesis testing without model-specific assumptions, proving versatile across both cross-sectional and time-series data. Gaussian process regression-based CIT (Flaxman et al. 2015) uses Gaussian process regression to model spatial and temporal dependencies, addressing both linear and nonlinear relationships in datasets characterized by their non-i.i.d. nature. The statistically equivalent signatures function (Lagani et al. 2017) expands CIT applications to correlated mixed data, analyzing datasets with both continuous and categorical variables.

However, these CITs generally demand a significant number of observations per subject for precision, a requirement not always met in datasets like ADNI. While the statistically equivalent signatures function broadens CITs to accommodate correlated mixed data, its restriction to linear relationships limits its applicability to datasets with complex, nonlinear dynamics. Repeated measurements data, with correlations within subjects across limited time points, pose distinct challenges these methods may not address effectively.

Within the context of previous CITs that assume i.i.d. data, the GCM provides a flexible framework for analyzing dependencies and relationships within complex data structures. This makes it an excellent candidate for extension to accommodate repeated measurements data. The strength of the GCM framework lies in its ability to integrate advanced regression techniques and machine learning algorithms. These tools have the potential to capture conditional dependencies between variables in non-i.i.d. settings—characteristics commonly observed in repeated measurements data. In the following, we aim to elaborate on the GCM framework, highlighting its potential in integrating novel regression methods to effectively address non-i.i.d. data.

3 Exploring the Generalized Covariance Measure Framework

The GCM for CI testing essentially involves regression analysis, where outcome variables $\mathbf{Y}^{(n)}$ and explanatory variables $\mathbf{X}^{(n)}$ are regressed against a set of conditioning variables $\mathbf{Z}^{(n)}$. This regression analyzes the relationships between the variables while accounting for the influence of conditioning factors. When the dimensions of X and Y (d_X and d_Y , respectively) are equal to one, the process focuses on evaluating the covariance of the residuals obtained from these regressions. This covariance, normalized and analyzed through the GCM, serves as the primary test statistic in evaluating CI. The continuity of the variables and the precision of the regression models in approximating conditional expectations are paramount for the successful application of the GCM to non-i.i.d. data (Shah & Peters 2020).

The flexibility and adaptability of the GCM make it a powerful tool for exploring and understanding the complexities in non-i.i.d. data structures across various domains (Hsieh et al. 2020; Q. Li et al. 2021). By allowing for the integration of domain-specific knowledge and the application of tailored regression techniques, the GCM offers a promising solution to the challenges posed by non-i.i.d. data. To more clearly explore an extension of its framework, we initially introduce the basic ideas of the GCM approach and its application to CI testing using different regression techniques.

3.1 Theoretical Foundations and Framework

This section initially discusses the statistical foundation and implications of the GCM with univariate variables X , Y , specifically focusing on the test statistic $T^{(n)}$, Type I error considerations, and the test's power.

The GCM's mathematical representation involves analyzing the conditional relationships among variables within a given distribution P of (X, Y, Z) . When the dimensions of X

3 Exploring the Generalized Covariance Measure Framework

and Y are equal to one, this is represented by the equations:

$$\begin{aligned} X &= f_P(Z) + \varepsilon_P, \\ Y &= g_P(Z) + \xi_P, \end{aligned}$$

where $f_P(z)$ and $g_P(z)$ denote the conditional expected values of X and Y given $Z = z$, respectively. $\varepsilon_{P,i}$ and $\xi_{P,i}$ are calculated as deviations from these conditional expectations for each data point $x_i - f_P(z_i)$ and $y_i - g_P(z_i)$ (Shah & Peters 2020).

To approximate these conditional expectations, the regression of $\mathbf{X}^{(n)}$ and $\mathbf{Y}^{(n)}$ against $\mathbf{Z}^{(n)}$ is estimated, resulting in $\hat{f}^{(n)}$ and $\hat{g}^{(n)}$. The product of the residuals for each data point is computed as follows:

$$R_i = \{x_i - \hat{f}(z_i)\}\{y_i - \hat{g}(z_i)\}.$$

This formulation leads to the construction of the test statistic $T^{(n)}$, a normalized sum of the residuals' products. To simplify the explanation, certain dependencies on n and P have been omitted. The test statistic $T^{(n)}$ is formulated as follows:

$$T^{(n)} = \frac{\sqrt{n} \cdot \frac{1}{n} \sum_{i=1}^n R_i}{\left(\frac{1}{n} \sum_{i=1}^n R_i^2 - \left(\frac{1}{n} \sum_{r=1}^n R_r \right)^2 \right)^{1/2}} =: \frac{\tau_N^{(n)}}{\tau_D^{(n)}}. \quad (3.1)$$

where test statistic $T^{(n)}$ is formulated as a ratio where the numerator, $\tau_N^{(n)}$, represents the normalized sum of the residuals' products, and the denominator, $\tau_D^{(n)}$, stabilizes this sum (Shah & Peters 2020).

In the forthcoming discussion, we address Type I error and the power of tests within the GCM framework for CI testing. For a detailed outline of the GCM's theoretical framework, as well as thorough discussions on Type I error and test power, please consult the Section A of the Appendix and the proofs conducted by Shah & Peters (2020).

It is essential that the test statistic's components $\varepsilon_i \xi_i$ be i.i.d., each with an expected mean of zero and finite variance, to ensure the validity of the test statistic's convergence to a normal distribution (Shah & Peters 2020):

$$\mathbb{E}_P(\varepsilon_i \xi_i) = \mathbb{E}_P\{\mathbb{E}_P(\varepsilon_i \xi_i | Y, Z)\} = \mathbb{E}_P\{\xi_i \mathbb{E}_P(\varepsilon_i | Z)\} = 0, \quad (3.2)$$

the test statistic $T^{(n)}$ asymptotically follows a standard normal distribution for univariate variables X and Y . This condition enables the determination of significance levels and

identification of Type I error, which is the probability of incorrectly rejecting a true null hypothesis H_0 .

The power of the test is its ability to correctly reject a false null hypothesis. Intuitively, the test statistic is designed as the normalized conditional covariance $\mathbf{cov}_P(X, Y|Z)$ between X and Y , given Z (Daudin 1980). This covariance is zero under the null hypothesis H_0 , which allows the test to be particularly responsive to alternative hypothesis where this covariance is not equal to zero (Shah & Peters 2020).

The mathematical representation is given by:

$$\mathbb{E}_P(\varepsilon_i \xi_i) = \mathbb{E}_P \mathbf{cov}_P(X, Y|Z), \quad (3.3)$$

where $\mathbf{cov}_P(X, Y|Z)$ is defined as $\mathbb{E}_P(XY|Z) - \mathbb{E}_P(X|Z) \mathbb{E}_P(Y|Z)$, quantifies the expectation of the product of residuals, which represent the discrepancies between observed values and their conditional expectations given Z . If H_0 holds true, indicating CI, this expected covariance should be zero, suggesting no shared variation between X and Y once Z is considered.

When extending the GCM to multivariate scenarios, the approach adjusts to account for complexities introduced by the multidimensional nature of variables X and Y , denoted as d_X and d_Y respectively. For each pair of variables, X_j from X and Y_k from Y , a test statistic $T_{jk}^{(n)}$ is computed for each combination of j and k , where j and k index the specific elements within the X and Y vectors, respectively. This process reflects an exhaustive examination of potential dependencies between each pair of variables across the two vectors (Shah & Peters 2020).

To synthesize the individual test statistics into a single comprehensive measure, S_n is defined as the maximum absolute value among all the $T_{jk}^{(n)}$ statistics:

$$S_n = \max_{j=1, \dots, d_X, k=1, \dots, d_Y} |T_{jk}^{(n)}|. \quad (3.4)$$

where S_n acts as an indicator of the strongest evidence of conditional dependence among all pairs of variables, revealing the presence of conditional dependence even if it is only observed between a few pairs. S_n is approximated by a $\hat{S}_n$, calculated under the condition that the collection of $T_{jk}^{(n)}$ statistics follows a mean-zero multivariate Gaussian distribution with a covariance matrix, which is derived from the data of $\mathbf{X}^{(n)}, \mathbf{Y}^{(n)}, \mathbf{Z}^{(n)}$ (Shah & Peters 2020).

The multivariate adaptation of GCM enables the CI testing in the datasets with multidimensional variables. By systematically computing test statistics for every variable

pair and aggregating them into a singular measure of significance, this approach ensures a thorough examination of potential dependencies.

From a theoretical standpoint, GCM emerges as an exemplary framework for integration with a wide array of regression techniques or machine learning algorithms, establishing it as a potent tool for CI testing. Practically, the selection of regression or machine learning methodologies to complement GCM typically depends on the dataset's specific characteristics and the analytical goals. The subsequent section will delve into how various regression techniques can be incorporated into the GCM framework.

3.2 Applying Regression Techniques within Framework

Within the scope of the GCM framework, the integration of diverse regression techniques significantly enhances its applicability and effectiveness in CI testing across various data settings. An overview of the current regression techniques within the GCM framework includes:

- Linear Regression and Generalized Additive Models (GAM): Essential tools for modeling both linear and nonlinear relationships between variables. Particularly, GAMs, implemented via the R package `mgcv` (S. Wood 2017), excel at capturing complex nonlinear trends without compromising interpretability (Lundborg et al. 2023).
- Boosted Regression Trees: Leveraging the `xgboost` package (T. Chen & Guestrin 2016), this approach adeptly addresses nonlinear interactions and intricate patterns within high-dimensional datasets (Shah & Peters 2020; Chalupka et al. 2018; Lundborg et al. 2023).
- Kernel Ridge Regression: Provides a method to tackle non-linear relationships within the GCM framework efficiently, thus expanding GCM's utility for various analytical challenges (Shah & Peters 2020).
- Sparse Linear Model with Least Absolute Shrinkage and Selection Operator (Lasso): Lasso regression extends GCM's reach to high-dimensional data analysis, facilitating optimal performance in scenarios where the conditioning variable Z exhibits a sparse inverse covariance structure (Ren et al. 2015; Shah & Peters 2020; Niu et al. 2023).

- Machine Learning Techniques: Incorporation of advanced algorithms such as GANs, enhancing GCM's analysis of complex data distributions (Shi et al. 2021; Lundborg et al. 2023).
- Regression Models with Functional Predictors: GHCM tailors GCM for functional random variables, accommodating complex data types like functional graphical models or regression models with functional predictors (Lundborg et al. 2022).

By incorporating a broad spectrum of regression methods, ranging from traditional models to state-of-the-art machine learning algorithms, the GCM framework establishes a comprehensive approach for CI testing. This approach can be specifically tailored to meet the unique demands of diverse datasets and analytical objectives. Consequently, the subsequent section will delve into the performance evaluation of GCM in conjunction with conventional GAM within the context of non-i.i.d. repeated measurements data.

3.3 Limitations in Analyzing Dependent Data

In order to demonstrate more clearly the limitations of GCM on non-i.i.d data, specifically within the context of repeated measurements, we provide a toy example. This example is designed to show the potential limitations posed by random effects within the GCM framework, particularly in terms of inflated Type I error rates under the null hypothesis. Below is a detailed description of the example setup and the rationale behind it.

We consider a linear example with random intercepts under the null hypothesis. Twenty subjects, each with two hundred repeated measurements, simulate a typical dataset for repeated measurements, illustrating a scenario where correlations within subjects are anticipated.

This example is specified as follows:

$$X_{it} = 1.5Z_{it} + \alpha_i + 0.3\varepsilon_{it}, \quad Y_{it} = 1.1Z_{it} + \beta_i + 0.3\xi_{it},$$

where Z_{it} represents the covariate for the i th subject at time t , sampled from a multi-variate standard normal distribution. α_i and β_i are random intercepts for each subject i , independently drawn from a normal distribution with mean zero and standard deviation of five. ε_{it} and ξ_{it} denote the noise components for each measurement, and follow a normal distribution with mean zero and variance one.

The aim of this example is to demonstrate how random effects (i.e., random intercepts α_i and β_i) can influence the result of GCM using GAM. The scatter plots in Figure 3.1

3 Exploring the Generalized Covariance Measure Framework

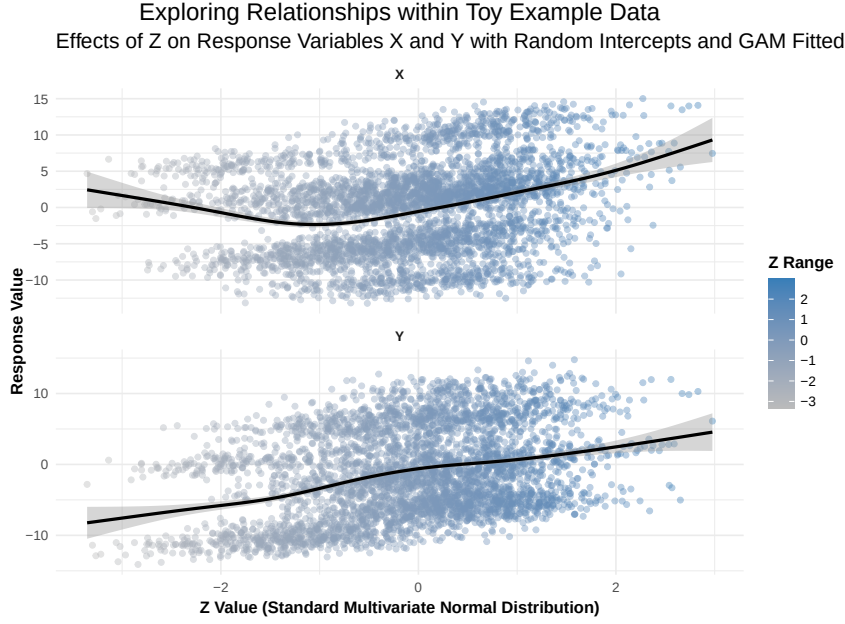


Figure 3.1: This figure showcases a scatterplot that investigates the relationships between variable Z and the response variables X and Y , as well as the fitting with GAM.

display an obvious linear relationship between the conditioning variable Z and each of the variables X and Y , respectively. This visualization confirms the linear models specified in the example. The observed linearity in the plots aligns with the null hypothesis H_0 , which posits that X and Y are CI given Z . Under H_0 , no significant association between X and Y should be detectable once Z is accounted for. The spread of points around the fitted line in each plot does not display any systematic pattern indicative of a departure from randomness. Additionally, the plots present the variation in intercepts across the population, emphasizing the impact of random effects.

Upon applying GCM, where GAM serves as the regression technique, the GCM yields a significantly low p-value (less than 0.01), suggesting the wrong rejection of the null hypothesis H_0 . This result indicates inflated Type I error rates, a critical issue when random effects are not adequately addressed in the model.

In Figure 3.1 GAM within the GCM framework can properly capture the linear relationship between the variables X , Y , and Z in our example. Theoretically, the flexibility of GAM in modeling such relationships should match the goals of GCM.

However, the residuals from GAM show a within-subject correlation in Figure 3.2. This discovery points to a violation of the fundamental assumptions underlying the GCM framework: $\varepsilon_i \xi_i$ be i.i.d., each with an expected mean of zero and finite variance. The visual evidence from Figure 3.2 provides a stark illustration of the limitations posed by the

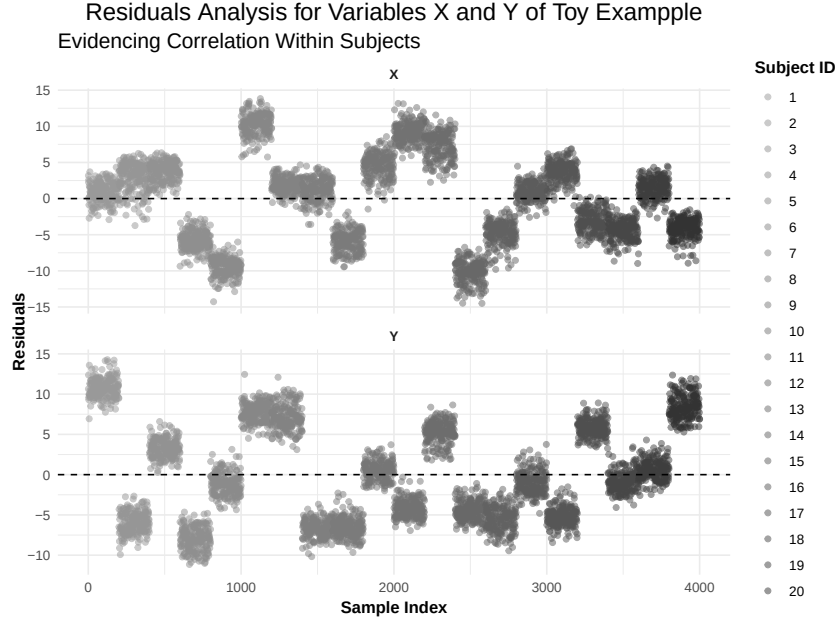


Figure 3.2: This figure displays the residuals obtained from GAM for variables X and Y , plotted against the sample index to visualize within-subject correlations explicitly. Each point denotes a residual value for a specific subject, distinguished by unique color coding, facilitating the observation of clustering patterns among subjects.

GCM when applied to non-i.i.d data. The GCM fails to deal with the correlation structure, specifically, wrongly reject the null hypothesis. In the context of CI testing, this means that the GCM might incorrectly suggest a dependency between X and Y conditional on Z , not because of any true association but due to the correlated structure of the data.

This toy example clearly demonstrates the limitations of GCM in the context of non-i.i.d. repeated measurements data. The conventional GCM, without suitable adjustments for random effects, misinterprets the correlation introduced by these random intercepts as indicative of a violation against H_0 . Such a misinterpretation emphasizes the need for an extended GCM approach, specifically designed for analyzing non-i.i.d. repeated measurements data. In the next chapter, we explore the commonly used methods for modeling repeated measurement data.

4 Advancements in Mixed Effects Models

The analysis of repeated measures data is crucial across various fields, including clinical research and environmental sciences, where observations on the same subjects are collected over multiple time points. To manage the complexity of non-i.i.d. repeated measurements, several models have been devised, each catering to different data structures and analytical needs. We begin with a general description of these models designed for repeated measurements.

Traditional methods like repeated measures analysis of variance (ANOVA) and Generalized Estimating Equations (GEEs) provide basic approaches for analyzing repeated measurements, though their utility may be constrained.

Repeated Measures ANOVA is particularly suited for analyzing continuous, normally distributed response variables. It allows for the evaluation of changes over time as well as differences between groups and its formula is given by:

$$Y_{it} = \mu + \lambda_i + \tau_t + (\lambda\tau)_{it} + \epsilon_{it},$$

where Y_{it} is the response for the i th subject at the t th time point, μ is the overall mean, λ_i the treatment effect, τ_t the time effect, $(\lambda\tau)_{it}$ the interaction between treatment and time, and ϵ_{it} the error term. However, its application is primarily effective in more straightforward, balanced experimental designs.

GEEs, on the other hand, focus on estimating average population parameters without fully specifying the probability model, according to the relationship:

$$E[Y_{it}|X_{it}] = g^{-1}(X_{it}\beta),$$

where $g^{-1}(\cdot)$ is the link function, offering greater flexibility in modeling the expected value of Y_{it} based on covariates X_{it} .

Multilevel Models (Goldstein et al. 1994; Maas & Snijders 2003) are good at exposing hierarchical structures and focusing on individual variations between observed time points

in favor of nested data structures:

$$Y_{it} = \pi_{0i} + \pi_{1i}\tau_{it} + \epsilon_{it},$$

where Y_{it} is the response for the i th individual at time t , τ_{it} represents the time or condition, π_{0i} and π_{1i} denote individual-specific intercepts and slopes, and ϵ_{it} is the residual error.

MEMs, however, distinguish themselves for their excellent adaptability in repeated measurements data. These model manage well the correlation of subject-specific data over time and can handle a wide range of data distributions, including missing data (Piepho et al. 2004; Hayat & Hedlin 2012; Davidian & Giltinan 2003; Barbosa & Goldstein 2000; Özgür Asar et al. 2015). MEMs integrate fixed and random effects alongside error terms, establishing a flexible analytical framework:

$$Y_{it} = f(X_{it}\beta) + Z_{it}b_i + \epsilon_{it}$$

where Y_{it} is the response at the t th time point for the i th subject, X_{it} and Z_{it} represent design matrices for fixed and random effects, respectively, β are the fixed effects, b_i signifies the random effect for the i th subject, f denotes a linear or nonlinear function, and ϵ_{it} is the residual error.

Among the introduced models, MEMs stand out for their potential flexibility and strength in the analysis of repeated measurements. These models offer an adaptable framework particularly suited to scenarios characterized by a limited number of measurements in our research question. In addition, MEMs is able to clarify the dependency structure typical of repeated measurements, containing both fixed effects (applicable to all observations) and random effects (which vary among subjects). Through the inclusion of random effects, MEMs afford more precise estimations of the regression functions for X and Y , given Z , thereby improving GCM's identification of conditional dependencies in repeated measurements data. The development of MEMs includes a wide range of types to meet the needs of different data complexities - from basic linear relationships to nonlinear and non-normally distributed data structures. The following briefly describes the development of MEMs models.

4.1 Recent Developments in Mixed Effects Models

Linear Mixed-Effect Models (LMEMs), as the foundational model within MEMs, are designed for managing fixed effects, which indicate consistent relationships across the

population, and random effects, which capture variability across subjects. The standard form of an LMEM is typically expressed as:

$$Y_{it} = X_{it}\beta + Z_{it}b_i + \epsilon_{it},$$

where Y_{it} denotes the response variable for the i^{th} subject at the t^{th} time point, X_{it} is the design matrix for fixed effects, β is a vector of the fixed effect coefficients, Z_{it} refers to the design matrix for random effects pertinent to the i^{th} subject, b_i is a vector of the random effect coefficients, and ϵ_{it} represents the error term, which is assumed to follow a normal distribution.

They are especially valuable for longitudinal data analysis, where they capture the inherent correlations within subjects over time. These models handle the variability both between and within groups, making them suitable for repeated measurements from the same subjects (Avilés 2001; D. Bates et al. 2014; J. C. Pinheiro & Bates 2001). LMEMs are best suited for data that conform to a linear model and require consideration of fixed and random effects.

When the relationships between predictors and response variables are nonlinear, Non-linear Mixed Effects Models (NMEMs) are the top models. They extend the LMEM to allow for complex relationships that cannot be captured by linear functions, making it applicable to a wider range of data structures. A typical NMEM can be expressed as:

$$Y_{it} = f(X_{it}, \beta) + Z_{it}b_i + \epsilon_{it}$$

In this expression, f represents a nonlinear function of the predictor X_{it} parameterized by the fixed effects β , which is what distinguishes it from LMEMs. The remaining terms are consistent with those outlined in the LMEM (Lindstrom & Bates 1990).

Generalized Linear Mixed Models (GLMMs) are a further extension that incorporates random effects into Generalized Linear Models (GLMs). These models can deal with data that has a hierarchical structure or does not follow to a normal distribution. Unlike LMEMs, GLMM can model a wide range of distributions in the exponential family, which allows the analysis of binary, count, and other types of response data (Nakagawa & Schielzeth 2013; Green & MacLeod 2016). A binomial GLMM example in Section B of the Appendix provided illustrates how these models can be used.

The formulation of a GLMM is given by:

$$g(\mu_{it}) = X_{it}\beta + Z_{it}b_i$$

where $g(\cdot)$ represents the link function, transforming the expected value of the response μ_{it} . Alternatively, the relationship $\mu_{it} = h(\eta_{it})$, where $h = g^{-1}$ denotes the inverse link function or the response function, is employed.

Besides these approaches, the newly proposed generalized additive mixed models (GAMMs) (Groll & Tutz 2011, 2012) are a combination of the features and strengths of LMEMs, NMEMs, and GLMMs. As a frontier approach, GAMMs can potentially benefit the analysis of repeated measures data for an extension of GCMs framework. Unlike LMEMs, which are constrained to linear associations, GAMMs can model complex non-linear trends within the data, capturing the subtleties of how predictors influence the response variables over time or across different conditions. Drawing from the strengths of GLMMs, GAMMs can analyze data that do not follow normal distribution. Distinct from GAMs, which are limited to fixed effects, GAMMs also incorporate random effects (C. Chen 2000).

The key advantage of GAMMs lies in their ability to manage correlations within observations grouped together, a common feature of repeated measurements data. By incorporating random effects, GAMMs could offer refined estimations of regression functions for X and Y conditioned on Z , thereby increasing the potential to identify true conditional dependencies in the data (Groll & Tutz 2011, 2012). Given these advantages, GAMMs become a promising option for extending the GCM framework, especially when dealing with non-i.i.d. data structures in context of repeated measurements. In the next sections, we take an in-depth look at GAMMs, explaining their model structure, and the theoretical foundations.

4.2 Applications of Generalized Additive Mixed Models

Initially, we explore a straightforward model incorporating a random intercept b_i to uncover the underlying principles of GAMM. This exploration will illustrate the inclusion of random effects and its relation to ridge penalties in the GAMM, as well as clarify how the basis functions can be formulated and selected within the framework of the model.

The simpler formula is provided as:

$$y_{it} = \beta_0 + f(x_{it}) + b_i + \epsilon_{it},$$

$$\text{where } \epsilon_{it} \underset{\text{ind}}{\sim} N(0, \sigma^2), \text{ and } b_i \underset{\text{ind}}{\sim} N(0, \sigma_b^2).$$

Fixed effects β_0 and $f(x_{it})$ represent the systematic component of the model, capturing

the relationship between the predictors and the response variable that is consistent across the population. Random intercepts b_i account for individual variability among subjects or groups, capturing the deviations of individual subject means from the population mean.

The random effects can be intuitively understood as analogous to shrinkage observed in ridge regression. Ridge regression serves as a technique to regularize regression coefficients, introducing a penalty on their magnitude to mitigate overfitting and multicollinearity by shrinking coefficients toward zero. This penalization forces the coefficients, particularly treats the random effects as parameters part, to shrink towards zero. However, implementing random effects via ridge penalties may be computationally demanding due to the inability to exploit the potentially sparse structure in many random effects (James et al. 2013).

The basis functions $B_k(x_{it})$ enable the modeling of the smooth function $f(x_{it})$. These functions expand the predictor variables into a space where the relationships between the variables can be modeled more flexibly:

$$f(x_i) = \sum_{k=1}^q B_k(x_{it})\beta_k.$$

where the smooth function $f(x_{it})$ is a combination of several basis functions B_k , each multiplied by a corresponding coefficient β_k .

This allows for a flexible, non-linear modeling of the predictor's effect (S. Wood 2017; James et al. 2013). In this case, random effects are similar to smooth functions with no basis functions that are entirely smooth, essentially acting as splines without smooth basis functions. Moreover, constraining the smoothness of these functions equates to assuming that the smooth coefficients β_k follow to a multivariate normal distribution with a mean of zero, functioning similarly to Gaussian noise as seen in LMEMs (Baayen et al. 2015).

These basis functions can range from polynomials to more complex splines, offering flexibility in modeling various types of data structures. For instance, polynomial functions of x can serve as basis functions, where each $B_k(x_{it})$ is simply a power of x_{it} :

$$B_k(x_{it}) = x_{it}^k,$$

this can be visualized in the polynomial regression model, allowing the modeling of non-linear trends as polynomials of degree d :

$$y_{it} = \beta_0 x_{it}^0 + \beta_1 x_{it}^1 + \beta_2 x_{it}^2 + \beta_3 x_{it}^3 + \dots + \beta_d x_{it}^d + \epsilon_{it},$$

4 Advancements in Mixed Effects Models

where $\epsilon_{it} \underset{\text{ind}}{\sim} N(0, \sigma^2)$, and $b_i \underset{\text{ind}}{\sim} N(0, \sigma_b^2)$.

The aforementioned straightforward model presents a simplified parametric model with a singular random intercept for each subject, meaning a homogeneous experimental experience across subjects. The model considers y_{it} as the response for subject i at time t , incorporating both fixed ($\mathbf{x}_{it}$) and random ($\mathbf{z}_{it}$) effects covariate vectors. The mean response $\mu_{it} = \mathbf{x}_{it}\beta + \mathbf{z}_{it}\mathbf{b}_i$ is modeled as a combination of these effects, reflecting the expected outcome given the fixed effects β and random effects $\mathbf{b}_i$. The model's parametric nature is rooted in its assumption of a linear combination of these effects:

$$\mu_{it} = \mathbf{x}_{it}\beta + \mathbf{z}_{it}\mathbf{b}_i,$$

with observations assumed to be conditionally independent with variances $\text{var}(y_{it}|\mathbf{b}_i)$. This setup helps understanding the foundational aspects of GAMM and their capacity to model data from repeated measurements effectively. In the following paragraphs details a more general model formulation within the GAMM framework, aiming to capture complex hierarchical and temporal data structures more efficiently.

The general framework of GAMM extends the initial linear parametric approach by incorporating nonparametric effects $\alpha_j(u_{itj})$ for each covariate $u_{it}^T = (u_{i1}, \dots, u_{itm})$, allowing for the modeling of complex, non-linear relationships through smooth functions. Consider y_{it} as the measurement for subject i at time point t , with $i = 1, \dots, n$ and $t = 1, \dots, T_i$, aggregated in the vector $\mathbf{y}_i^T = (y_{i1}, \dots, y_{iT_i})$. The covariate vector for fixed effects is denoted by $\mathbf{x}_{it}^T = (1, x_{it1}, \dots, x_{itp})$, and for random effects by $\mathbf{z}_{it}^T = (z_{it1}, \dots, z_{itq})$. This semiparametric model combines the linear parametric effects with the additive nonparametric effects and random effects, creating a flexible framework for detailed data analysis (Groll & Tutz 2011, 2012):

$$g(\mu_{it}) = \mathbf{x}_{it}^T\beta + \sum_{j=1}^m \alpha_j(u_{itj}) + \mathbf{z}_{it}^T\mathbf{b}_i = \mathbf{X}\beta + \mathbf{B}\alpha + \mathbf{Z}\mathbf{b}, \quad (4.1)$$

where g is a link function connecting the mean response μ_{it} to the linear combination of fixed $\mathbf{X}\beta$, nonparametric $\mathbf{B}\alpha$, and random effects $\mathbf{Z}\mathbf{b}$. $\mathbf{x}_{it}^T\beta$ constitutes the linear parametric term, including an intercept, $\sum_{j=1}^m \alpha_j(u_{itj})$ embodies an additive term with unspecified influence functions, and $\mathbf{z}_{it}^T\mathbf{b}_i$ encompasses cluster-specific random effects, with $\mathbf{b}_i$ adhering to a normal distribution (Taavoni & Arashi 2024). An alternative expression is $\mu_{it} = h(\eta_{it})$, with $h = g^{-1}$ as the inverse link function (Groll & Tutz 2011, 2012).

The GAMM's non-linear modeling capability lies in the estimation of smooth functions

$(\alpha_j(u))$ through basis functions, typically regression splines, which are widely utilized in estimating additive structures (S. Wood 2017; Ruppert et al. 2003; Wand 2000). This approach, grounded in the flexibility of splines, especially B-splines, offers a robust framework for estimating the unknown smooth functions $\alpha_j(\cdot)$ that represent the effects of covariates on the response variable. The choice of basis functions and their degree d is critical in defining the model's capacity to approximate complex non-linear trends accurately.

B-splines of degree d are particularly noted for their ability to provide a smooth, flexible curve that can adapt to various types of data structures. The formulation is defined as:

$$\alpha_j(u) = \sum_{i=1}^k \alpha_{ji} B_{ji}^j(u; d),$$

where $B_{ji}^d(u; d)$ represents the i -th B-spline basis function for the j -th covariate, captures this adaptability. The degree d of the spline affects the smoothness and flexibility of the curve, with higher degrees offering more flexibility but requiring more parameters to be estimated. $\boldsymbol{\alpha}_j^T = (\alpha_{j1}, \dots, \alpha_{jk})$ contains the coefficients for the j -th smooth function, which directly affects the shape and direction of the estimated curve.

The equations are an illustration of the integration of these splines into the GAMM framework:

$$g(\mu_{it}) = \mathbf{x}_{it}^T \boldsymbol{\beta} + \mathbf{B}_1^T(u_{it1}) \boldsymbol{\alpha}_1 + \dots + \mathbf{B}_m^T(u_{itm}) \boldsymbol{\alpha}_m + \mathbf{z}_{it}^T \mathbf{b}_i.$$

where each term $\mathbf{B}_j^T(u_{itj}) \boldsymbol{\alpha}_j$ represents the contribution of the j -th covariate's non-linear effect on the response (Groll & Tutz 2011, 2012).

The fitting for these GAMMs is often a choice between Restricted Maximum Likelihood (REML) (Patterson & Thompson 1971, 1974) and Penalized Quasi-Likelihood (PQL) (Breslow & Clayton 1993; Lin & Breslow 1996; Breslow & Lin 1995). REML is widely recognized as a preferred method of choice for fitting GAMMs because of its robustness and accuracy in estimating variance components (Smyth & Verbyla 1996). REML is adept at providing reliable estimates of variance components, a feature that becomes increasingly valuable in small sample scenarios where the risk of biased estimates is heightened (Kenward & Roger 1997). One of the significant advantages of REML is its ability to correct for the degrees of freedom consumed in estimating fixed effects, thus leading to less biased estimates across the model. Within the GAMM framework, the application of REML to the selection of smoothing parameters is critical to ensure that

the model optimizes the balance between accurately fitting the structure of the data and preventing overfitting, which is essential for models that include additive terms (Reichert 1993; Meza et al. 2009). PQL offers a different approach, especially useful for models with response variables following distributions within the exponential family. (Breslow & Clayton 1993; Lin & Breslow 1996; Breslow & Lin 1995). For a more detailed explanation, we describe the PQL applied to GAMM in the Section C of the Appendix. However, PQL has been observed to yield biased estimates for variance components, particularly in models dealing with binary outcomes or Poisson data with low counts. This bias undermines the model’s accuracy, especially in smaller datasets (S. Wood et al. 2017; S. Wood & Wood 2015; S. Wood 2017).

Model selection in GAMMs involves balancing complexity with data fidelity. Traditional techniques like Akaike Information Criterion (AIC), Bayesian Information Criterion (BIC), or Generalized Cross-Validation (GCV) are employed to compare models based on their likelihood, aiding in the identification of a model that best fits the data while avoiding overfitting (S. Wood 2017; S. N. Wood 2011). However, these criteria may struggle with models that include complex random effects due to the computational challenges presented by the dependency on observed data (H. Wang et al. 2007; Rohloff et al. 2022). To prevent overfitting, which can occur when too many effects are included, a simpler MEM that are less likely to overfit the data (Ibrahim et al. 2011; Zou 2006; Fan & Li 2001). The techniques like Smoothly Clipped Absolute Deviation (SCAD) (Fan & Li 2001) and Adaptive Least Absolute Shrinkage and Selection Operator (ALASSO) (Zou 2006) introduce penalties for additional parameters, and ensure that the selected model captures the essential features of the data without being overly complex. A more detailed description can be found in the Section D of the Appendix

In the exploration of GAMMs and their fitting, we investigate the advantages of REML over PQL, with a particular emphasis on REML’s robustness in variance component estimation and its strength in dealing with small sample sizes and complex random effects structures. This discussion emphasizes the importance of selecting an appropriate fitting method based on the model’s objectives and the data characteristics, to ensure accurate and reliable estimation. Next, we discuss the specific software implementation of MEMs, including GAMMs. In the subsequent section, we delve into the specific software implementations of MEMs, including GAMMs.

4.3 Software for Mixed Effects Model Implementation

The implementation of MEMs in R is facilitated by several comprehensive packages, each designed to meet specific modeling needs. Notably, the `lme4` package (D. M. Bates 2010) and `nlme` package (J. Pinheiro et al. 2013) packages provide foundational tools for fitting linear and non-linear mixed-effects models, respectively, while `mgcv` package (S. Wood & Wood 2015) offers advanced capabilities for modeling complex random effects structures through a wide range of smoother.

Among these tools, the `gamm4` package (S. Wood et al. 2017) package is specifically designed for fitting GAMMs, combining the modeling capabilities of `mgcv` with the mixed-model functions of `lme4` package. This synergy leverages the numerical stability of `lme4` for underlying model fitting, making `gamm4` particularly suited for complex data scenarios that challenge other packages. Based on these advantages, `gamm4` shows its great capabilities when dealing with non-i.i.d. random effects or a large number of random coefficients, where its model-fitting approach proves most effective (S. Wood et al. 2017).

For the implementation of GAMM in the thesis, the choice of `gamm4` package (version 0.2-6) (S. Wood et al. 2017) was motivated by its robust fitting functions and its adeptness at selecting smoothness via REML for Gaussian additive models and Maximum Likelihood for other distributions. Specifically, we utilize the `gamm` fitting functions within `gamm4`. The package is able to deal with the characteristics of our repeated measurements data, with a particular focus on accuracy and reliability, making it ideal for fitting specified GAMMs.

The discussion on fitting methods and software implementations sets the stage for the next chapter, where we aim to extend the GCM framework to adapt repeated measurements. This extension aims to leverage the strengths of GAMMs, particularly their flexibility in modeling complex relationships and variability within the data, to enhance the applicability of the GCM framework to a wider range of research protocols. In the following chapter, we will detail an extension of GCM framework for repeated measures, exploring how GAMMs can be effectively utilized to capture the complexity of repeated measures data.

5 Innovating with an Extended Generalized Covariance Measure Framework

In this chapter, we detail our extended framework EGCM for CI testing with repeated measurements data. First reflecting on the toy example previously discussed in Section 3.3, we illustrated the limitations in the traditional GCM framework when dealing with non-i.i.d. repeated measurement data. In the GCM framework, the utilization of GAM for regression analysis led to an erroneously low p -value (less than 0.01). This suggested a false rejection of the null hypothesis H_0 , potentially due to the framework's inability to adequately model the data's underlying structure and correlations, especially those introduced by the repeated measurements.

In the extension, known as the EGCM, we leverage the strengths of GAMMs to model the intricate relationships and dependencies within the data. Our implementation involved using `gamm4` package in R to model GAMMs, specifying `(1|subject)` to denote a random intercept that varies by subject. This specification allows for the consideration of both the relationship between X and Y with Z and the correlation among individual subjects within the analysis.

Conversely, the application of EGCM, which integrates GAMMs, produced a p -value of 0.4205. This value represents the same expectation under the null hypothesis, indicating a failure to reject H_0 correctly.

A critical limitation in the GCM approach is the presence of correlation among residuals within subjects when dealing with repeated measurements as in Figure 3.2 showed. The EGCM framework, by integrating GAMMs, offers a robust solution to this issue. In contrast to the GCM approach, where residual correlation within subjects is a notable concern, the EGCM framework exhibits no apparent correlation in within-subject residuals in Figure 5.1. The random distribution of residuals around this line, with no discernible pattern tied to subject, underscores the efficacy of the GAMM in addressing the intricate structure characteristic of repeated measurements.

Based on the results of this simple example, it is evident that EGCM provides a more sophisticated and effective framework for analyzing repeated measurements data compared to the GCM. In the following sections, we will delve deeper into the specific settings of

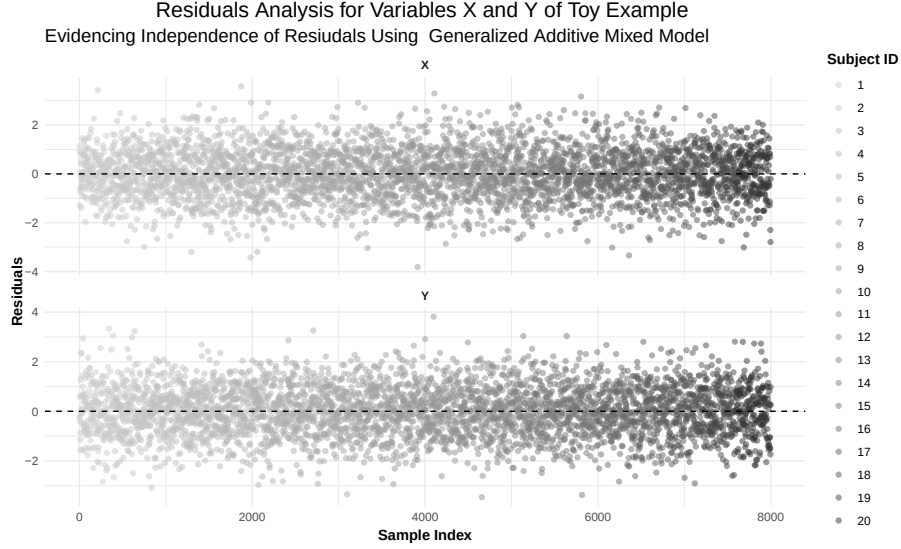


Figure 5.1: This scatterplot illustrates the residuals from GAMM of variables X and Y within a toy example. The residuals are plotted against the sample index, with individual subject IDs differentiated by various shades of grey. The consistent dispersion of residuals around the horizontal zero line across all subject IDs suggests that the model has successfully accounted for individual variations, exhibiting no apparent trends or systematic deviations within the data of any subject.

EGCM and outline the testing process in detail, demonstrating its application in univariate and multivariate contexts and the theoretical foundation of the EGCM framework.

5.1 Theoretical Foundations of the Extended Framework

The EGCM framework aims to enrich the CI testing procedure by incorporating an advanced regression modeling technique GAMM to estimate the expected values of variables X_{it} and Y_{it} , given the influence of covariates Z_{it} with f and g being distinct monotonic differentiable link function. This approach significantly enhances the accuracy and reliability of the estimation process, particularly in datasets with repeated measurements and non-linear characteristics.

5.1.1 Integrating an Enhanced Regression Method for Dependent Data

The expected value for variable X_{it} , for instance, is modeled as:

$$\mu_{X,it} = E[X_{it} | \mathbf{b}_{X,i}, \mathbf{Z}_{it}, \mathbf{r}_{X,it}],$$

which is represented by the following equation:

$$f(\mu_{X,it}) = \mathbf{Z}_{it}^T \boldsymbol{\beta}_X + \sum_{j=1}^m \alpha_{X,j}(u_{X,itj}) + \mathbf{r}_{X,it}^T \mathbf{b}_{X,i}. \quad (5.1)$$

Similarly, for Y_{it} , the expected value $\mu_{Y,it} = \mathbb{E}[Y_{it} | \mathbf{b}_{Y,i}, \mathbf{Z}_{it}, \mathbf{r}_{Y,it}]$ is captured by:

$$g(\mu_{Y,it}) = \mathbf{Z}_{it}^T \boldsymbol{\beta}_Y + \sum_{j=1}^m \alpha_{Y,j}(u_{Y,itj}) + \mathbf{r}_{Y,it}^T \mathbf{b}_{Y,i}. \quad (5.2)$$

where linear parametric components, represented by $\mathbf{Z}_{it}^T \boldsymbol{\beta}_X$ and $\mathbf{Z}_{it}^T \boldsymbol{\beta}_Y$, these components account for the fixed effects associated with variables X and Y , respectively, offering a baseline estimation that incorporates known Z covariates. Additive nonlinear terms $\alpha_{X,j}(u_{X,itj})$ and $\alpha_{Y,j}(u_{Y,itj})$ introduce flexibility into the model by incorporating smooth functions that capture nonlinear relationships within the data, enhancing the model's adaptability to complex data structures. Random effects components, captured by $\mathbf{r}_{X,it}^T \mathbf{b}_{X,i}$ and $\mathbf{r}_{Y,it}^T \mathbf{b}_{Y,i}$, these components model the variability not explained by fixed effects or the included covariates, addressing the intra-subject variability and dependencies across time points.

Applying GAMM within the EGCM framework contributes to the careful analysis of repeated measures data, which allows uncovering the complexities of relationships and correlations that exist within and between subjects over time. This approach improves not only the accuracy of the estimates but also contributes to a deeper understanding of the underlying data structure, which in turn improves the robustness of the CI testing. The general procedure for fitting GAMMs, as outlined in the Algorithm 1, involves a systematic approach that includes model specification, selection of appropriate basis functions for smooth components, estimation of parameters, and comprehensive interpretation of the results. This structured procedure ensures that the analysis remains robust and insightful, providing a solid foundation for CI testing in complex datasets.

5.1.2 Extended Test Statistic for Univariate X and Y

Building on the foundational principles of the GCM, which utilized residuals derived from the expected values $f(z) = \mathbb{E}[X|Z = z]$ and $g(z) = \mathbb{E}[Y|Z = z]$, the EGCM further adapts this concept to suit the complexity inherent in repeated measurements data. This adaptation is made possible by incorporating GAMMs, which offer a refined approach to estimating expected values, therefore, the residuals are redefined in a way that recognizes

Algorithm 1 GAMMs fitting procedure using `gamm4`

- 1: **Input:** Data (Y_{it}, X_{it}, Z_{it}) for $i = 1, \dots, n$ and $t = 1, \dots, h_i$.
- 2: **Define** model structure for X_{it} and Y_{it} , specifying appropriate link functions and incorporating fixed, random, and smooth additive effects:

$$f(\mu_{X,it}) = \mathbf{Z}_{it}^T \boldsymbol{\beta}_X + \sum_{j=1}^m \alpha_{X,j}(u_{X,itj}) + \mathbf{r}_{X,it}^T \mathbf{b}_{X,i},$$

$$g(\mu_{Y,it}) = \mathbf{Z}_{it}^T \boldsymbol{\beta}_Y + \sum_{j=1}^m \alpha_{Y,j}(u_{Y,itj}) + \mathbf{r}_{Y,it}^T \mathbf{b}_{Y,i}.$$

- 3: **Select** basis functions for non-parametric effects (e.g., B-splines or natural cubic splines) to accurately model the smooth components.
- 4: **Estimate** model parameters via `gamm4`, focusing on the REML maximization for both X_{it} and Y_{it} . Adjust the penalty terms accordingly to mitigate the risk of overfitting.
- 5: **Refine** the model iteratively, utilizing model selection criteria (e.g., AIC, BIC, GCV) to identify the optimal model configuration, paying close attention to the potential for overfitting, especially with complex random effects structures.
- 6: **Validate** the fitted model, assessing its performance through diagnostic checks and, if necessary, applying techniques (e.g., SCAD, ALASSO) to further enhance model accuracy.

Return: Fitted functions $\hat{f}(\mu_{X,it})$ and $\hat{g}(\mu_{Y,it})$, alongside a comprehensive analysis of model performance.

Remark: By leveraging the `gamm4` package in R, this procedure ensures a robust and efficient fitting of GAMMs, particularly for models with intricate random effects structures, enhancing the fidelity of CI testing in repeated measurements data.

and considers the non-i.i.d. characteristics of the data.

The refined estimation of expected values for X_{it} and Y_{it} , denoted as $f(\mu_{X,it})$ and $g(\mu_{Y,it})$, respectively. By leveraging the advanced modeling potential of GAMMs, the EGCM not only maintains the integrity of the original GCM's principles but also extends its applicability to datasets characterized by the non-i.i.d. nature of repeated measurements.

The revised formula for calculating the product of residuals, which now accommodates the non-i.i.d. data, is given by:

$$R_{it} = \{x_{it} - f(\mu_{X,it})\}\{y_{it} - g(\mu_{Y,it})\}, \quad (5.3)$$

where $f(\mu_{X,it})$ and $g(\mu_{Y,it})$ symbolize the expected values of X_{it} and Y_{it} , as estimated by GAMMs in Algorithm 1.

The extended test statistic $T^{(n,M)}$ in EGCM is recalculated as a normalized sum of these revised residual product terms:

$$T^{(n,M)} = \frac{\sqrt{M} \cdot \frac{1}{M} \sum_{i=1}^n \sum_{t=1}^{h_i} R_{it}}{\left(\frac{1}{M} \sum_{i=1}^n \sum_{t=1}^{h_i} R_{it}^2 - \left(\frac{1}{M} \sum_{i=1}^n \sum_{t=1}^{h_i} R_{it} \right)^2 \right)^{1/2}}, \quad (5.4)$$

where $T^{(n,M)}$ incorporates an understanding that not all subjects in a repeated measurements study contribute the same number of observations. M , to represent the total number of measurements across all subjects, is defined as the sum of individual measurement counts for each subject, mathematically expressed as $M = \sum_{i=1}^n h_i$. Each subject i is associated with a unique count of measurements, denoted by h_i .

The numerator of $T^{(n,M)}$ calculates the average product of residuals R_{it} across all subjects and their respective measurements. This is achieved by summing the products of residuals for each time point t for each subject, from the first measurement to their last, h_i , measurement, and then averaging these sums across all subjects. The denominator normalizes this average by computing the square root of the variance of these product terms. Specifically, it subtracts the square of the average of the residuals' product from the average of the squared residuals' product, for each subject from their first to their last measurement. This variance calculation ensures the test statistic accounts for variability in the residuals' products across all observations.

The EGCM is summarized in Algorithm 2. This methodological improvement benefits a more refined evaluation of CI by determining if the derived test statistic exceeds a predetermined critical threshold corresponding to a selected significance level α . This

approach is especially advantageous for exploring CI testing of interrelationships between variables within datasets marked by both fixed and random effects, as well as potential nonlinear associations.

When the GAMMs effectively account for the influence of Z_{it} on both Y_{it} and X_{it} with REML, the residuals, ε_{it} and ξ_{it} , are expected to be i.i.d., each with a zero mean and finite variance (Patterson & Thompson 1971, 1974; Reichert 1993; Smyth & Verbyla 1996; Meza et al. 2009). Based on the CLT and the theorems in the GCM framework (Shah & Peters 2020) as discussed in Section 3.1, the extended test statistic $T^{(n,M)}$ converges to a normal distribution.

Algorithm 2 EGCM for Univariate X and Y

Input: Data $(X_{it}, Y_{it}, Z_{it})_{i=1, t=1}^{n, h_i}$, Total measurements $M = \sum_{i=1}^n h_i$, significance level $\alpha \in (0, 1)$.

Options: Use GAMM methods for each of the regressions as outlined in Algorithm 1.

1: Estimate GAMM-based residuals:

- (i) Regress X_{it} on Z_{it} and Y_{it} on Z_{it} using GAMM to obtain estimated functions $\hat{f}(\mu_{X,it})$ and $\hat{g}(\mu_{Y,it})$.
- (ii) For each observation pair (i, t) , compute residuals:

$$\varepsilon_{it} = X_{it} - \hat{f}(\mu_{X,it}), \quad \xi_{it} = Y_{it} - \hat{g}(\mu_{Y,it}).$$

2: Compute the product of residuals for each (i, t) :

$$R_{it} = \varepsilon_{it} \cdot \xi_{it}.$$

3: Formulate the EGCM test statistic $T^{(n,M)}$, by computing the normalized sum of R_{it} across all (i, t) :

$$T^{(n,M)} = \frac{\sqrt{M} \cdot \frac{1}{M} \sum_{i=1}^n \sum_{t=1}^{h_i} R_{it}}{\sqrt{\frac{1}{M} \sum_{i=1}^n \sum_{t=1}^{h_i} R_{it}^2 - \left(\frac{1}{M} \sum_{i=1}^n \sum_{t=1}^{h_i} R_{it} \right)^2}}.$$

4: Reject the null hypothesis H_0 , if $T^{(n,M)} > z_{1-\alpha}$.

Note: This procedure leverages the GAMM fitting method to account for complex dependencies and non-linear interactions within the data, enhancing the precision of CI testing in the context of repeated measurements. It utilizes the total number of measurements M across all subjects to accurately normalize and compute the test statistic, ensuring a robust analysis that reflects the true variability of the dataset.

5.1.3 Aggregated Test Statistic for Multivariate X and Y

In a multivariate setting, the goal is to test for CI between vectors $\mathbf{X}$ and $\mathbf{Y}$ given $\mathbf{Z}$. To achieve this, we compute univariate EGCM test statistics, $T_{jk}^{(n,M)}$, for each pair of variables (X_j, Y_k) where $j = 1, \dots, d_X$ and $k = 1, \dots, d_Y$. This pairwise approach allows for the nuanced exploration of dependencies between each component of $\mathbf{X}$ and $\mathbf{Y}$, given $\mathbf{Z}$. The culmination of this process is the aggregation of these pairwise statistics into a singular multivariate test statistic, denoted as $S^{(n,M)}$.

The multivariate test statistic, $S^{(n,M)}$, is defined as the maximum absolute value among all individual $T_{jk}^{(n,M)}$ statistics:

$$S^{(n,M)} = \max_{j,k} |T_{jk}^{(n,M)}|. \quad (5.5)$$

This aggregation strategy effectively captures the strongest evidence of conditional dependence across all variable pairs, thereby offering a comprehensive measure of CI within the multivariate framework (Shah & Peters 2020). The asymptotic distribution of $S^{(n,M)}$ under the null hypothesis of CI is approximated through Monte Carlo simulations as discussed in Section 3.1. We summarize this process of EGCM with multivariate X and Y in Algorithm 3.

This analysis aims to explain the theoretical viability of EGCM, suggesting that the robustness of GAMMs in analyzing repeated measurements data might be enhanced. By aiming to capture data dependencies and non-linearities, GAMMs might contribute to improving EGCM's compliance with the GCM assumptions. This could ensure the asymptotic behavior of the test statistic. Applying the EGCM may provide a promising framework for CI testing, particularly within datasets that exhibit repeated measurements and non-i.i.d. characteristics. In the forthcoming section, we will discuss how these theoretical considerations are translated into practical diagnostic tools for model fitting and CI testing.

5.2 Critical Role of Residual Analysis in Conditional Independence Testing

This section emphasized that in the context of CI testing, especially when employing regression-based testing methods such as the GCM framework and its variants, the role of residual analysis is critical. Specially, the application of GCM and its extended version EGCM are based on the analysis of the normalized covariance of residuals to test

Algorithm 3 EGCM for Multivariate X and Y

Input: Data $(X_{it}, Y_{it}, Z_{it})_{i=1, t=1}^{n, h_i}$, where $X_{it}, Y_{it} \in \mathbb{R}^{d_X}, \mathbb{R}^{d_Y}$ respectively, represent the multivariate responses at time t for subject i , with dimensions d_X and d_Y , and significance level α .

Options: Use GAMM methods for regression as outlined in Algorithm 1.

- 1: **Estimate GAMM-based residuals** for each component of X and Y , conditional on Z :
 - (i) For each component X_j and Y_k , regress onto Z using GAMM to obtain estimated functions $\hat{f}_j(\cdot)$ for X_j and $\hat{g}_k(\cdot)$ for Y_k .
 - (ii) Compute residuals $\varepsilon_{it,j} = X_{it,j} - \hat{f}_j(Z_{it})$ for X_j and $\xi_{it,k} = Y_{it,k} - \hat{g}_k(Z_{it})$ for Y_k .
- 2: **Compute the product of residuals** $R_{it,jk} = \varepsilon_{it,j} \cdot \xi_{it,k}$ for all component pairs (j, k) .
- 3: **Formulate the test statistic** $S^{(n,M)}$ as the maximum absolute value among all individual test statistics $T_{jk}^{(n,M)}$ derived from $R_{it,jk}$:

$$S^{(n,M)} = \max_{j=1, \dots, d_X, k=1, \dots, d_Y} |T_{jk}^{(n,M)}|.$$

- 4: **Estimate the empirical distribution** $\hat{S}_{1-\alpha}^{(n,M)}$ is approximated using Monte Carlo simulations as a mean-zero multivariate Gaussian distribution with a sample covariance matrix $\hat{\Sigma}$.
- 5: **Reject** the null hypothesis H_0 , if $S^{(n,T)} > \hat{S}_{1-\alpha}^{(n,T)}$.

Note: This algorithm integrates multivariate responses into the EGCM framework, leveraging GAMMs to enhance the test's adaptability to repeated measurements. The use of Monte Carlo simulations accommodating the intricacies of multivariate cases.

for CI among variables. From a previous toy example, we observed that GCM could mistakenly reject the null hypothesis due to significant correlations found in the residuals after regression. However, the EGCM, which utilizes improved GAMMs for non-i.i.d. data, successfully did not reject the null hypothesis, accurately identifying the CI state in the data. This distinction is attributed to the lack of noticeable correlation in the residuals depicted in the EGCM residual plot. Specifically, ensuring that the residuals after regression possess an i.i.d. mean and finite variance can lead to more accurate CI testing.

For a clearer understanding of the discussion on the application of GCM and EGCM to non-i.i.d. data, two remarks are given:

Remark 1. *In the context of non-i.i.d. data, where the standard i.i.d. assumption of the GCM framework is violated, as long as the residuals ε_{it} and ξ_{it} meet criteria $\mathbb{E}(\varepsilon_{it}\xi_{it}) = 0$, the GCM framework remains viable for CI testing. These residuals are obtained from the regressions of X on Z and Y on Z , respectively. This implies that even under the violation of i.i.d. assumption, if these residuals, specifically their products $\varepsilon_{it}\xi_{it}$, are i.i.d. with mean of zero and finite variance. The GCM framework can still effectively test for CI among the variables X , Y , and Z .*

Remark 2. *When dealing with non-i.i.d. data in the GCM framework, the residuals ε_{it} and ξ_{it} obtained through the standard regression methods not meet the condition $\mathbb{E}(\varepsilon_{it}\xi_{it}) = 0$, leading to biased CI testing results. However, incorporating GAMM tailored for non-i.i.d. data into the framework, thus forming an EGCM, ensures that the residuals through the improved regression method fulfill this i.i.d. condition with mean zero and finite variance. The EGCM can accurately test for CI in such situations where the GCM framework might fail.*

These remarks underscore the importance of analyzing residuals in regression-based tests. It not only helps in enhancing the accuracy of CI testing but also prevents biased results of CI testing when analyzing complex data structures. This practice is underscored by a body of research (Bagnato & Punzo 2010; Q. Chen et al. 2020; Flaxman et al. 2015; Sen & Sen 2013). For instance, a CIT based on residual similarity for linear structural equation models was introduced (H. Zhang et al. 2023).

Based on this discussion, we strongly recommend incorporating an analysis of residuals in the CI testing process as summarized in the following CI testing process. This refined procedure emphasizes the importance of a thorough residual analysis in the process of CI testing within the GCM and EGCM frameworks. By incorporating both visual and

statistical assessments of residuals, and adjusting the regression model as necessary, we can more accurately interpret the results of CI testing. Therefore, integrating this additional step of residual analysis is crucial, designed to improve the methodological precision and effectiveness of GCM and EGCM when navigating the intricacies of data with inherent dependencies and non-linear relationships.

Analyzing Residuals upon Rejecting the H_0 in GCM and EGCM Frameworks

- 1: **Regress X and Y onto Z using regression methods.**
 - 2: **Analyze residuals for independence:**
 - (a) Plot residuals ε_{it} and ξ_{it} over their respective indexes separately to visually assess any patterns, trends, or structures.
 - (b) Conduct statistical tests for autocorrelation and independence, such as the Durbin-Watson test for residuals, to quantitatively examine their independence.
 - 3: **If correlated residuals are detected:**
 - (a) Explore more sophisticated regression methods that can account for the observed correlation structure among observations. This might involve using methods detailed in the EGCM framework, which are designed to handle non-i.i.d. repeated measurements data.
 - (b) Utilize scatterplots and partial correlation plots of X , Y , and Z to further investigate potential causal relationships and to ensure that the model accurately reflects the underlying data structure.
 - 4: **Interpret results with caution:**
 - Be mindful of the fact that the validity of the GCM and EGCM results heavily relies on the assumption of i.i.d. residuals with mean zero and finite variance. Correlated residuals can introduce a bias in CI testing.
 - Acknowledge that a failure to reject H_0 may not conclusively indicate CI among variables X , Y , and Z . It is possible that correlated residuals might obscure true relationships.
-

In the following chapter, we undertake empirical experiments designed to compare the performance of the GCM and the EGCM specifically within the context of non-i.i.d. repeated measurements data. This practical evaluation will aim to substantiate the theoretical advantages of EGCM posited in earlier sections, particularly its robustness and adaptability in effectively handling repeated measurements. Through this comparative analysis, we seek to demonstrate the efficacy of EGCM as a superior method for CI testing, offering insights into its potential for broader application in CI testing involving non-i.i.d. repeated measurements data.

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In this chapter, we provide a detailed exploration through simulation studies and empirical analysis in real ADNI data, focusing on the applicability of the GCM and EGCM in the area of repeated measurements. The simulations aim to examine the models' adeptness in accurately identifying CI across diverse and complex data structures that mirror the intricacies encountered in real-world repeated measurements. Furthermore, the application to the ADNI dataset clarifies the critical role of the EGCM framework in enhancing our understanding of early AD diagnostics. By testing CI between structural MRI markers and a precise criterion for AD diagnosis using amyloid positron emission tomography (PET) given traditional cognitive assessment scores, this study demonstrates the potential of MRI markers as predictive tools for AD, emphasizing the importance of advanced EGCM in revealing the relationship between structural changes in the brain volumes and cognitive function. This investigation not only highlights improvements from the EGCM in CI testing but also demonstrates its real-world applicability in providing deeper insights into the detection of neurodegenerative diseases.

6.1 Validating the Extended Framework through Simulation Studies

In this section, we detail simulation studies conducted to assess the performance of the original GCM and its extended version EGCM, specifically when dealing with complex, non-i.i.d. repeated measurements data:

(1) Models with random intercept:

This initial scenario evaluates the methods' performance in complex, non-linear relationships. The EGCM demonstrated a pronounced improvement in identifying CI amidst non-linear dynamics and random intercepts, starkly outperforming the GCM. This underscores the EGCM's superior adaptability to non-linear complexities.

(2) Models with random slope:

Extending the complexity, this simulation incorporates both random intercepts and slopes to mirror the complexities in real-world datasets, including randomly introduced missing values. EGCM outperforms GCM even when models are misspecified

only to include random intercepts, demonstrating EGCM's robustness and accuracy in identifying CI across diverse scenarios.

(3) Synthetic ADNI dataset:

Focusing on challenges prevalent in real medical datasets, such as incomplete data and varying observation counts, a synthetic dataset inspired by the real-world ADNI dataset is generated. The simulation confirmed the EGCM's effectiveness in handling complex, medically relevant datasets, emphasizing its potential for broad applications in medical research.

These simulations are executed across 100 iterations to evaluate Type I error rates and power, and leverage various R packages, including `mgcv` for GAM (S. Wood 2017; S. Wood & Wood 2015), `gamm4` for GAMMs (S. Wood et al. 2017), `GeneralisedCovarianceMeasure` for the GCM (Shah & Peters 2020).

6.1.1 Models with Random Intercept

The section on models with random intercepts offers a detailed exploration into the performance of GCM and EGCM across various data dynamics, extending the exploration beyond linear relationships to include complex interactions such as square, cubic, hyperbolic tangent, and negative exponential functions (K. Zhang et al. 2012). This variability in simulation runs mirrors the diverse empirical data relationships, enabling a comprehensive evaluation of the methods under different realistic data scenarios.

The simulation design considers varying numbers of subjects $n \in \{20, 50, 100, 200\}$ and measurements per subject $h_i \in \{3, 5, 10, 20, 50, 100\}$, allowing for a thorough examination of the methods' performance across different data configurations. In our investigation, we emphasize the importance of sufficient sample sizes for achieving reliable estimates of conditional means. Shah & Peters (2020)'s work suggested that the GCM can effectively control Type I errors and maintain robust power even with sample sizes of only up to 400.

This simulation is structured around two main scenarios, each with its unique setup to mimic different research situations:

- **Scenario (a): Simple Correlation Structures.**

In the initial scenario, the data generation process focuses on creating simple correlation structures between variables. This is achieved by simulating covariate Z_{it} from a multivariate normal distribution with a specified covariance matrix. The relationships between the variables are defined through different functions f and g ,

which are applied to Z_{it} to generate X_{it} and Y_{it} , respectively:

$$X_{it} = f(Z_{it}) + 0.3\varepsilon_{it},$$

$$Y_{it} = g(Z_{it}) + bX_{it} + 0.3\xi_{it},$$

where Z_{it} is drawn from a multivariate normal distribution with mean 0 and a specific covariance structure Σ , where the covariance Σ_{it} is 0.5 for $i \neq t$ and the variance Σ_{ii} is 1. This setup aims to simulate typical within-subject correlations observed over time.

- **Scenario (b): Incorporating Random Intercepts.**

The second scenario introduces complexity by including random intercepts for each subject. This involves generating Z_{it} from a standard multivariate normal distribution but adds random intercepts α_i and β_i for each subject. These intercepts are drawn from a normal distribution with mean zero and standard deviation five:

$$X_{it} = f(Z_{it}) + \alpha_i + 0.3\varepsilon_{it},$$

$$Y_{it} = g(Z_{it}) + \beta_i + bX_{it} + 0.3\xi_{it},$$

these setups are designed to reflect the complexities of real-world data, including the effect of unobserved individual differences (random intercepts) and the variability of measurements within subjects over time.

For the analysis, GAM and GAMM are employed to handle the data from both scenarios. Specifically, the `mgcv` package is used for GAM, while the `gamm4` package is used for GAMM to incorporate random intercepts. The GAMMs for X and Y are specified as $X \sim s(Z) + (1|\text{subject})$ and $Y \sim s(Z) + (1|\text{subject})$, respectively. $s(Z)$ denotes the application of a smoothing function to the covariate Z , and $(1|\text{subject})$ represents a random intercept that varies by subject.

The simulation's primary aim is to compare the EGCM's performance against the GCM for Type I error control and power. With $b = 0$, simulating the null hypothesis where X and Y are CI given Z , both GCM and EGCM are found in Figure 6.1 to satisfactorily control Type I errors in scenario (a), which features simpler correlation structures. This suggests that both models conform to theoretical expectations, accurately reflecting the data's correlation structure and effectively controlling the Type I error. Furthermore, when introducing different covariance matrix for Z from the `nlme` package in R, GCM

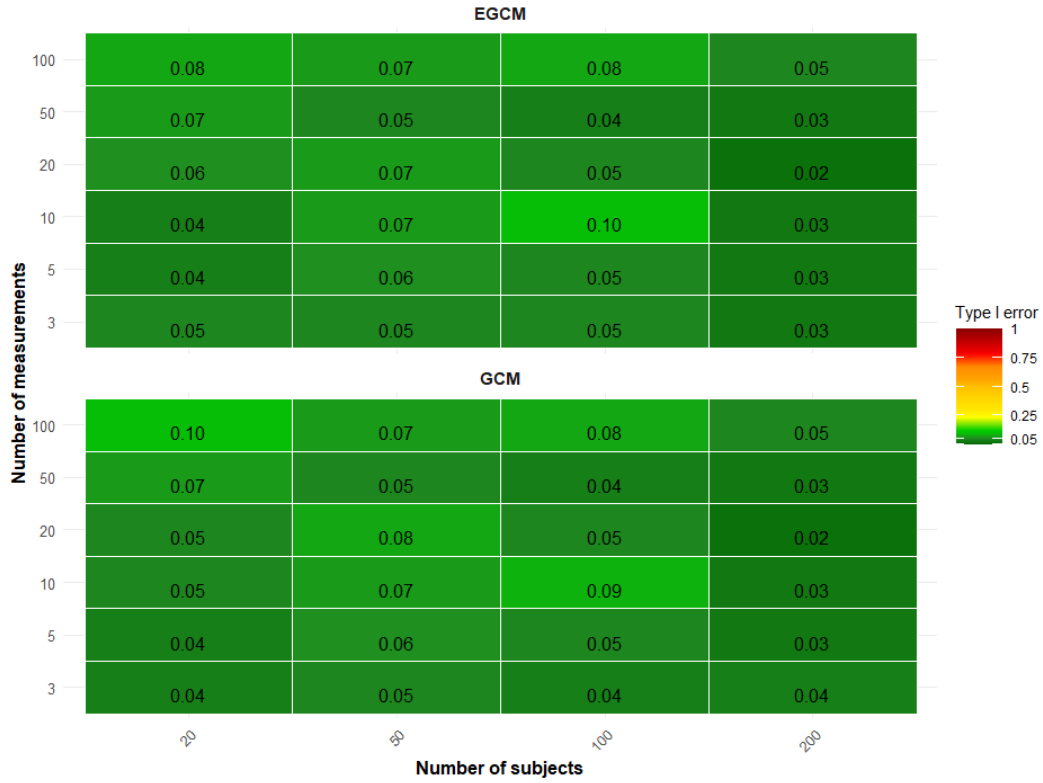


Figure 6.1: The heatmap's upper section showcases the EGCM, while its lower section illustrates the GCM. Lighter green shades signify error rates approaching 0.05, symbolizing superior performance in Type I error control. In scenario (a), characterized by simpler correlation structures, both GCM and EGCM demonstrate satisfactory Type I error control, as evidenced by the majority of cells aligning with the target error rate.

continued to exhibit reliability in maintaining Type I error control.

In the scenario (b), the Figure 6.2 reveals significant differences. In scenario (b), which involves random intercepts, the GCM struggles to maintain Type I error control, particularly as the number of measurements increases. In contrast, the EGCM exhibits remarkable resilience in the face of complexity, as evidenced by the predominance of green shades, especially in cases with a larger number of measurements and subjects. This underscores EGCM's superior ability to manage the intricacies of CI testing that involves random intercepts and repeated measurements.

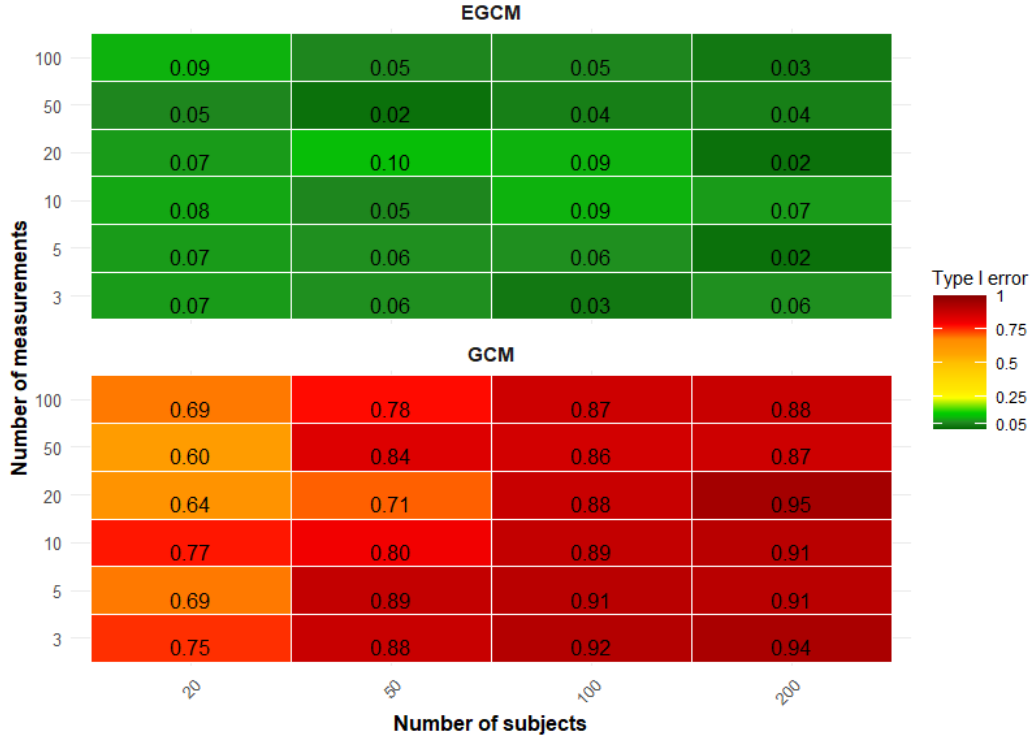
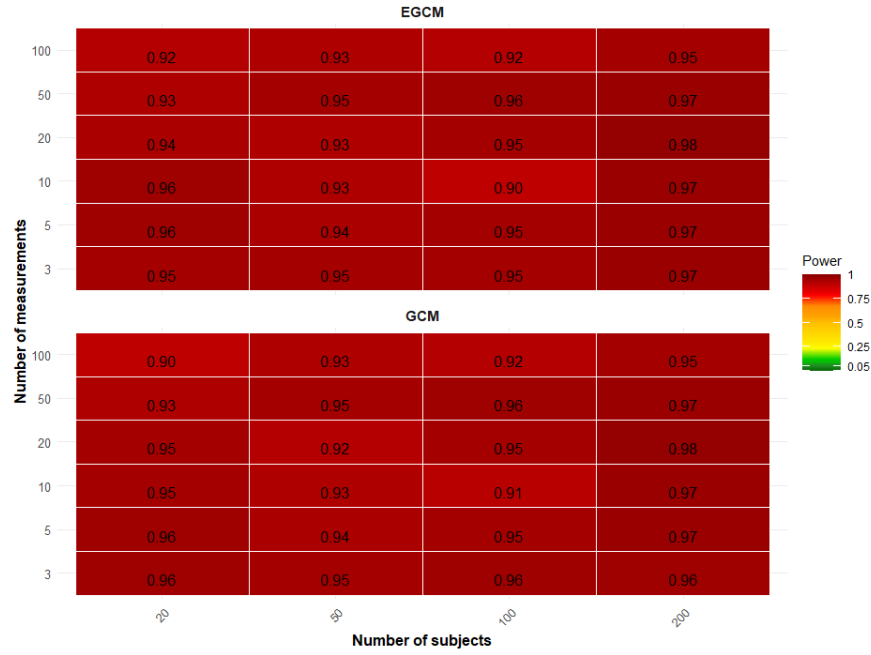


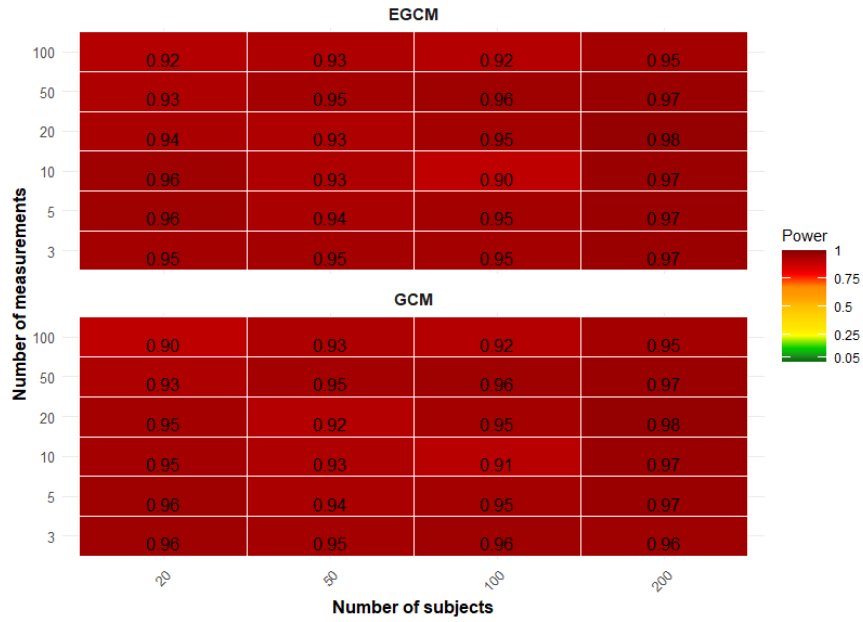
Figure 6.2: This heatmap delineates the disparate performance of the GCM and the EGCM in controlling Type I errors within a scenario characterized by random intercepts. The upper portion of the heatmap illustrates EGCM's efficacy in maintaining lower Type I error rates, depicted through varying shades of green. Conversely, the lower portion, representing GCM, exhibits notably higher Type I error rates, as indicated by shades of orange and red.

The introduction of $b = 0.2$ aims to simulate the alternative hypothesis where X and Y are not CI given Z , providing insights into the tests' power. Figure 6.3 showcases the power results for scenarios (a) and (b), respectively, illustrating that both GCM and EGCM effectively reject the null hypothesis with enough power. This finding highlights the potential conservative nature of GCM in indicating conditional dependence among X , Y , and Z , particularly in more complex data settings, compared to EGCM. EGCM exhibits potentially a more pronounced ability to discern statistical significance levels and reject the null hypothesis where appropriate.

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- (a) This heatmap delineates the power of the GCM and the EGCM in scenarios (a). Across the heatmap, cells with red tones predominate, indicating high power values approaching 1. This pattern suggests that both GCM and EGCM possess robust power in rejecting the null hypothesis of no CI between X and Y , given Z .



- (b) This heatmap delineates the power of the GCM and the EGCM in scenarios (b). In this comparison, the EGCM shows a consistent red tone across the heatmap, indicating that it consistently maintains high power. The GCM displays a somewhat varied range of power, yet still maintains high power overall.

Figure 6.3: Comparative Evaluation of Power in Scenarios (a) and (b)

6.1.2 Models with Random Slope

The section on models adding random slopes details an advanced simulation setup. This simulation incorporates more complex real-world conditions by introducing subject-specific random effects, including both intercepts and slopes, and simulating the occurrence of missing data. The aim is to understand how well each model performs in terms of controlling Type I error rates and maintaining test power under various conditions, including correct model specifications and misspecifications.

The simulation varies the number of subjects n across several levels (20, 50, 100, 200, 300, 400) while keeping the number of measurements per subject h first fixed at ten. This approach helps assess the models' robustness against changes in sample size. For each subject, random intercepts α_i and β_i and slopes RE_{sx} and RE_{sy} are generated to introduce individual variability. The intercepts are drawn from a normal distribution with mean zero and standard deviation five, while the slopes are drawn from a normal distribution with mean zero and standard deviation 1.5. The independent variable Z is generated for each observation from a standard normal distribution $N(0, 1)$. Reflecting common research challenges, missing values are introduced into the X , Y , and Z variables with a missing probability of 0.2. This step simulates the occurrence of data loss due to various factors, such as non-response or technical errors during data collection, making our simulation more reflective of practical analytical scenarios.

The generation formulas for X and Y incorporate the random effects to simulate subject-specific deviations:

$$X_{it} = \alpha_i + f(Z_{it}) + RE_{sx}Z_{it} + 0.3\varepsilon_{it},$$

$$Y_{it} = \beta_i + g(Z_{it}) + RE_{sy}Z_{it} + bf(Z_{it}) + 0.3\xi_{it}.$$

where ε_{it} and ξ_{it} are error terms independently drawn from a normal distribution $N(0, 1)$. RE_{sx} and RE_{sy} represent the random slopes for X and Y , respectively.

The simulation study was designed to assess the performance of EGCM in comparison with GCM across various conditions, including different subject sizes and both correct and incorrect model specifications. The investigation focused on the models' ability to control Type I error rates under the null hypothesis and their power in detecting true associations under the alternative hypothesis.

Under the correct model specification, which includes both random intercepts and slopes $(1 + Z/\text{Subject})$, EGCM demonstrates superior Type I error control compared to GCM

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across all subject sizes in Figure 6.4. Type I error rates for EGCM are significantly lower, indicating a robust capacity to avoid false positives under the null hypothesis ($b = 0$). In terms of power, EGCM outperforms GCM under the alternative hypothesis ($b = 0.2$), with EGCM achieving higher power across different subject numbers. This shows EGCM's enhanced ability to correctly identify true associations when they exist.

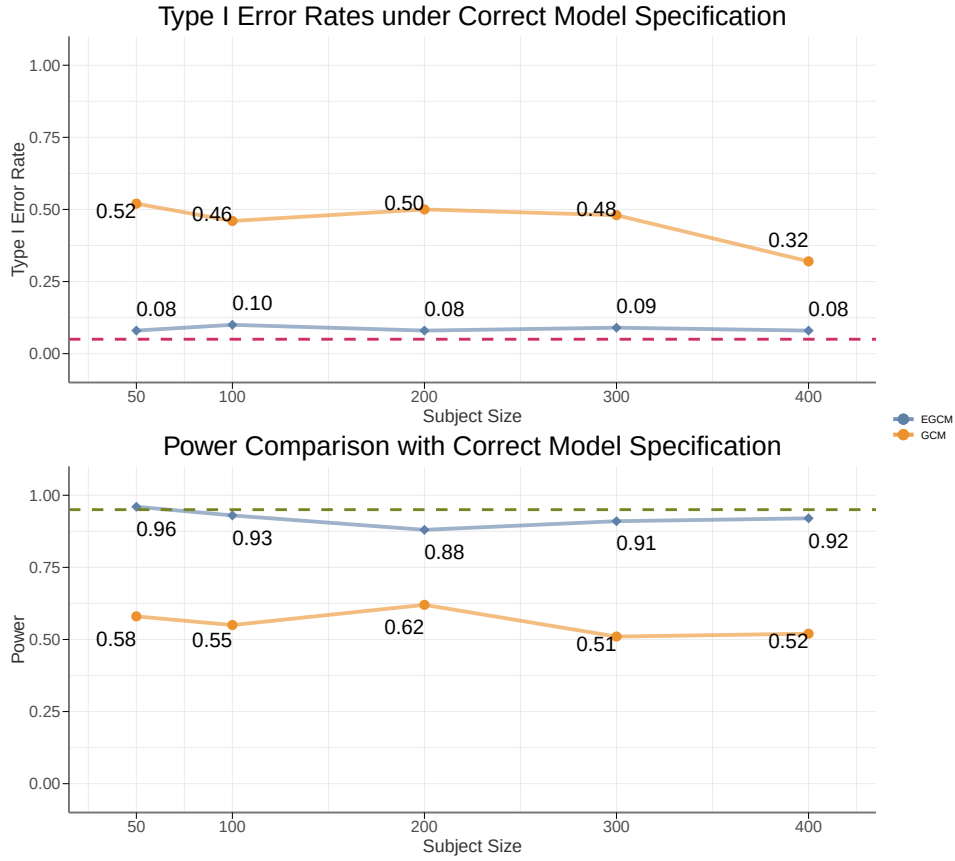


Figure 6.4: This figure consists of two graphs, depicting the performance of the EGCM and the GCM when correctly specifying models that incorporate both random intercepts and slopes. The upper graph illustrates Type I error rates, while the lower graph demonstrates the statistical power across varying sizes of subjects. Notably, EGCM surpasses GCM in controlling Type I errors under correct model specifications, for all sizes of subject populations. This enhanced performance highlights EGCM's robustness and reliability in CI testing.

Even when the model is misspecified to only include random intercepts ($1/Subject$), EGCM still outperforms GCM. Despite the misspecification, EGCM maintains lower Type I error rates and higher power compared to GCM in Figure 6.5, demonstrating its robustness and accuracy in identifying CI.

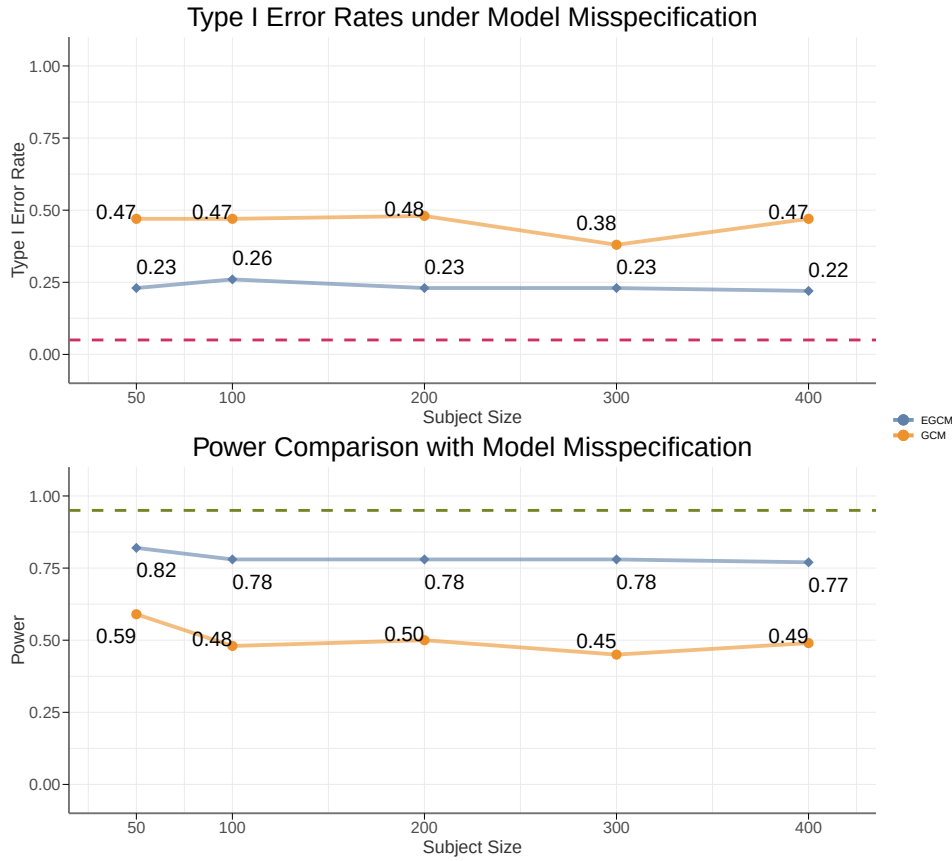


Figure 6.5: This composite figure elucidates the comparative performance on Type I error rates and statistical power amidst model misspecification for both the EGCM and the GCM. Despite model misspecifications that encompass only random intercepts, EGCM surpasses GCM in both Type I error management and statistical power. This accentuates EGCM's robustness and precision in CI testing, rendering it a more dependable choice, even under suboptimal model conditions.

The performance of GCM, which approached a 0.5 probability in Type I error control and power across different settings, suggests a near-random determination in rejecting or accepting the null hypothesis. This indicates a significant limitation of GCM, especially in the presence of repeated measurements and random effects. EGCM's consistent performance, even under model misspecification, underscores its superior methodology for CI testing. It shows that EGCM is more adept at handling the complexities of data with random effects, providing a more reliable and sensitive measure for testing CI. Particularly in a data-generating process with strong random effects, EGCM's efficacy remains as good as when the model is correctly specified. Under such conditions, EGCM maintains Type I error rates below 0.1 and power above 0.85, even when GCM's Type I errors and power fluctuate around 0.5. More detailed results and data settings can be found in

Section F of the appendix. These findings support the comment that EGCM provides a more sensitive and accurate measure for CI testing, even in the most basic setting of repeated measurements with random intercepts. The insight guides our evaluation of the performance of EGCM and GCM in real ADNI data, assuming the simplest model of repeated measurements with random intercepts.

The simulation study conclusively demonstrates EGCM’s superior performance over GCM in controlling Type I errors and maintaining test power under various conditions. This includes scenarios with correct model specifications (including random intercepts and slopes) and misspecifications (considering only random intercepts). These results highlight EGCM’s robustness and accuracy in conditional independence testing, making it a more reliable tool for analyzing complex empirical data. This insight is critical for guiding future research and analysis, particularly in fields where understanding the nuances of conditional relationships is paramount. In our experimental exploration, the control of Type I error under the null hypothesis presents a more pronounced challenge for the GCM compared to the power under the alternative hypothesis. This finding has led us to primarily focus on exploring the control of type I error terms in subsequent analyses.

6.1.3 Synthetic ADNI Dataset

In this section, we delve into the performance of the EGCM in scenarios that closely mimic the complexities of real-world data, drawing inspiration from the ADNI dataset. We address the challenge of model overfitting, especially pertinent in high-dimensional data or complex model designs. To mitigate this, we explore the use of a simpler model with random intercept. Our analysis focuses on the efficacy of EGCM in navigating non-i.i.d. conditions within a framework of repeated measurements, simulating scenarios reflective of the ADNI dataset’s structure. This includes incorporating random intercepts and a variable number of measurements per subject, to emulate real data complexities. Previous simulations indicated a challenge for the GCM in controlling Type I error rates under the null hypothesis compared to EGCM. By examining both univariate and bivariate variable configurations (Shah & Peters 2020), we focus on the discussion of the control of Type I error terms, and contribute to the understanding of EGCM’s potential for CI testing within repeated environments in medical research.

Synthetic data is generated in the following steps: (1) Utilize real-world data from the ADNI study via `Adnimerge` package (ADNI Team 2023). (2) Determine minimum, maximum, and proportion values (number of measurements per subject). (3) Apply the method of moments for continuous variables to transform data into a normal distribution

shape. (4) Adjust the transformed data back to its original range and round to the nearest integer (Handels et al. 2024). We generate the number of measurements per subject h_i by sampling from a normal distribution with a mean of 6.73 and a standard deviation of 4.75, with values rounded to the nearest whole number and constrained from one up to twenty-four measurements per subject. This approach simulates a data environment reflecting the variability and missingness inherent in real-world ADNI datasets.

To thoroughly assess EGCM and GCM across a wide range of scenarios, our simulation expands to include diverse subject counts $n \in \{200, 600, 1000, 1200, 1500\}$. This allows for a detailed examination of dataset size impacts on the performance of GCM and EGCM, especially in terms of Type I error control. The data was generated according to the following specifications:

- Univariate Configuration:

$$\begin{aligned} X_{it} &= f_1(Z_{1it}) + f_2(Z_{2it}) + \alpha_i + 0.3\varepsilon_{it}, \\ Y_{it} &= g_1(Z_{1it}) + g_2(Z_{2it}) + \beta_i + 0.3\xi_{it}. \end{aligned}$$

where f_1 , f_2 , g_1 , and g_2 are functions representing a mixture of linear and non-linear relationships, selected randomly to capture the variability inherent in empirical data. The terms α_i and β_i denote random intercepts introducing within-subject variability, while ε_{it} and ξ_{it} are error terms.

- Bivariate Configuration: Variables X , Y , and Z were expanded to incorporate a bivariate framework, enhancing the simulation's complexity to align with real-world data scenarios:

$$Z_{1it}, Z_{2it} \sim \mathcal{N}(0, 1), \quad \alpha_{1i}, \alpha_{2i}, \beta_{1i}, \beta_{2i} \sim \mathcal{N}(0, 5^2).$$

$$\begin{aligned} X_{1it} &= f_1(Z_{1it}) + f_2(Z_{2it}) + \alpha_{1i} + 0.3\varepsilon_{1it}, \\ X_{2it} &= f_1(Z_{1it}) + f_2(Z_{2it}) + X_{1it} + \alpha_{2i} + 0.3\varepsilon_{2it}, \\ Y_{1it} &= g_1(Z_{1it}) + g_2(Z_{2it}) + \beta_{1i} + 0.3\xi_{1it}, \\ Y_{2it} &= g_1(Z_{1it}) + g_2(Z_{2it}) + Y_{1it} + \beta_{2i} + 0.3\xi_{2it}. \end{aligned}$$

forming a multivariate system that allows for a deeper exploration of the models' performance under more intricate conditions.

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The results of the simulations, illustrated in Figure 6.6, reveal that EGCM exhibits superior performance compared to GCM, particularly in maintaining Type I error rates close to the 0.05 level. This finding underscores EGCM's effectiveness in controlling Type I error, a critical aspect where GCM struggles, especially in the face of data complexities mimicking real-world conditions.

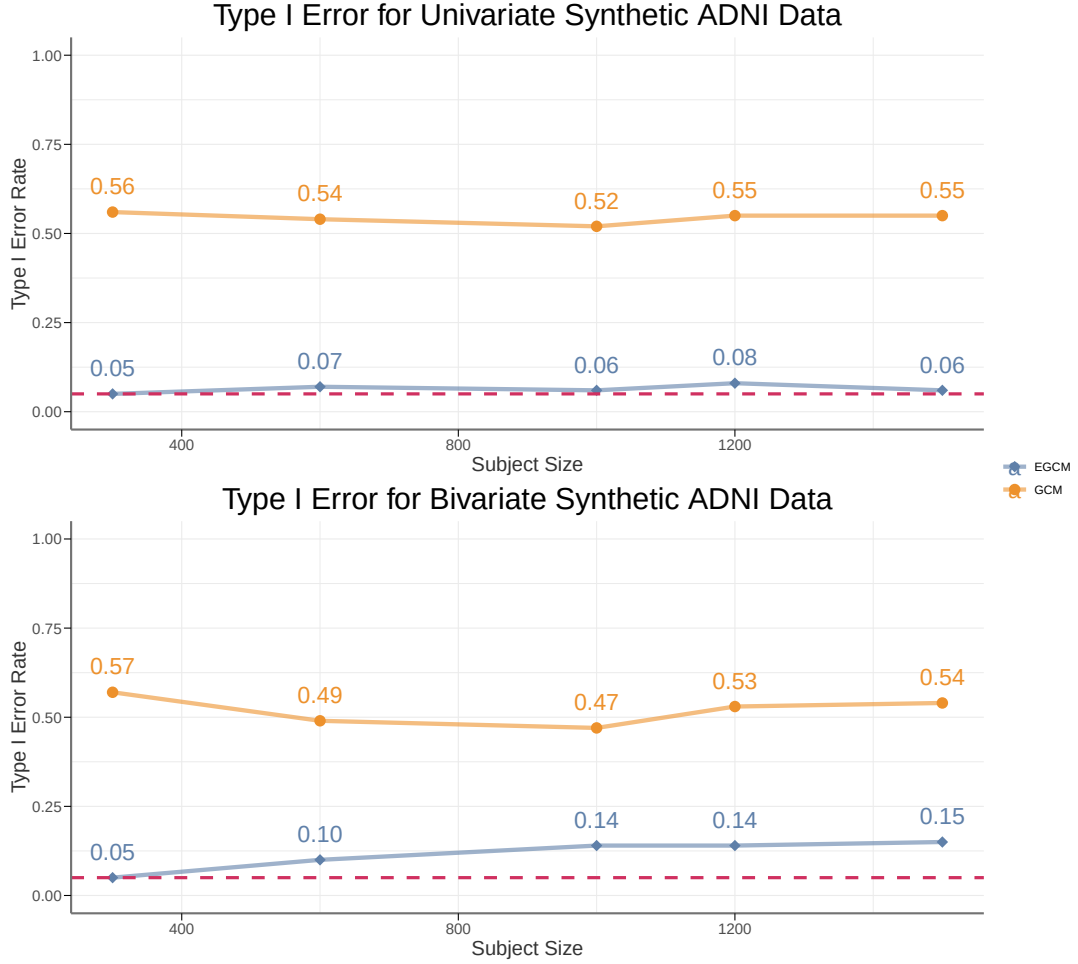


Figure 6.6: This figure showcases two line graphs illustrating Type I error rates for univariate and bivariate synthetic ADNI data across various subject sizes. The upper graph illustrates the univariate data analysis, whereas the lower graph details the bivariate data analysis. In both representations, the blue line representing the EGCM closely adheres to the 0.05 (indicated by a dotted pink line), demonstrating effective Type I error control. In contrast, the GCM, depicted by an orange line, exhibits Type I error rates significantly approaching a 0.5. Such effectiveness of the EGCM underscores its appropriateness for the complex data analyses commonly encountered in ADNI research.

The simulation study cautiously suggests that EGCM may offer valuable enhancements over traditional GCM, particularly in dealing with non-i.i.d. data within repeated measure-

ments frameworks. The inherent challenges posed by real medical datasets, such as missing data, highlight EGCM’s potential as a refined method for conditional independence testing in complex situations.

Looking forward, there is potential for further evolution within the GCM framework to accommodate more complex dependence structures and employ advanced model selection techniques to mitigate the risk of overfitting. This future exploration could include integrating random slopes, examining interactions among various random effects, or incorporating more sophisticated models, possibly leveraging neural network architectures. The aim is to expand the applicability of the EGCM framework to a broader range of data structures while balancing model complexity and interpretability, ensuring the methodology remains relevant and effective in addressing the nuanced challenges of empirical research.

6.2 Case Analysis Using Real Alzheimer’s Disease Neuroimaging Initiative Data

In this section, we explore the CI testing of structural MRI markers within the AD diagnostic process, given traditional cognitive assessment scores, using GCM and its extension, the EGCM. This analysis uses ADNI data (ADNI Team 2023) to potentially enhance our understanding of AD beyond the insights provided by cognitive tests alone. The problem of interest in this investigation focuses on the predictive potential of structural MRI markers for AD progression, compared to traditional cognitive assessments. Structural MRI markers such as hippocampal volume, overall brain volume, ventricular enlargement, and medial temporal lobe atrophy are closely associated with AD’s neuropathology. These markers, particularly hippocampal atrophy and ventricular enlargement, provide early prognostic information and are critical for diagnosing AD at an early stage, potentially improving patient outcomes through timely intervention.

The study emphasizes the diagnostic potential of key structural MRI markers such as hippocampal volume, ventricular enlargement, and medial temporal lobe atrophy, which are closely linked to the neuropathology of AD (Popuri et al. 2020; Khatri & Kwon 2022; Leandrou et al. 2018; Olsen et al. 2017). These markers, particularly hippocampal atrophy and ventricular enlargement, provide valuable prognostic information from the early stages of AD. The reduction in volume in the entorhinal cortex, surrounding the hippocampus, is another significant indicator of early AD diagnosis, underscoring the utility of these MRI markers in enhancing early diagnostic efforts (Dickerson et al. 2001; Thompson et al. 2005; Zhou et al. 2016). These MRI markers, by reflecting specific and early-stage alterations in

the brain's structure, offer essential insights into the early diagnosis and monitoring of AD's progression.

Cognitive assessments like the MMSE, the Montreal Cognitive Assessment (MoCA) and the Clinical Dementia Rating Scale Sum of Boxes (CDR-SB) are pivotal in diagnosing AD, assessing cognitive functions such as memory and attention (van Maurik et al. 2017). Despite their importance, the limitations of these tests motivate the exploration of additional diagnostic indicators like structural MRI markers (Silverberg et al. 2011; Loewenstein et al. 2018; Mortamais et al. 2017; Hobart et al. 2013).

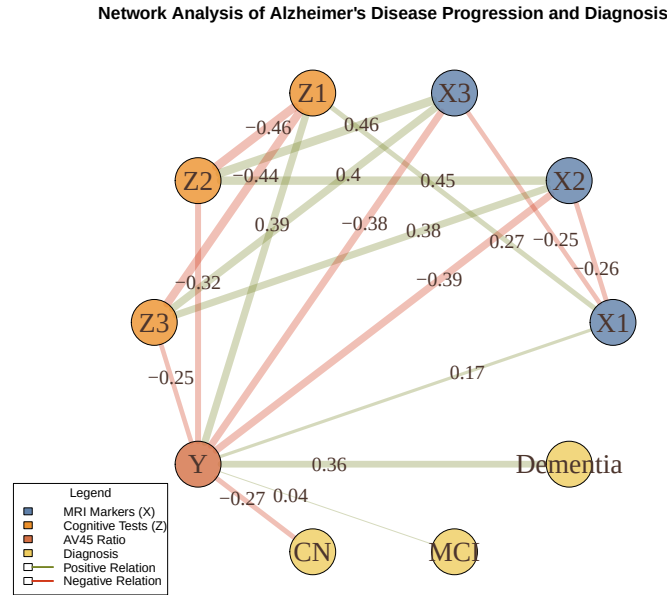


Figure 6.7: This network diagram delineates the interrelations among MRI markers (ventricles X_1 , hippocampus X_2 , entorhinal cortex X_3), cognitive test scores (MOCA Z_1 , MMSE Z_2 , CDR-SB Z_3), and the AV45 ratio (Cortical Grey Matter to Whole Cerebellum), in relation to the diagnoses of cognitively normal (CN), mild cognitive impairment (MCI), and dementia. The line thickness signifies the strength of correlation, with green lines illustrating positive associations and red lines depicting negative ones. Numerical labels on the connections represent the correlation coefficients between the corresponding variables.

The main focus of our study is to evaluate the CI of MRI markers with early AD detection given cognitive test outcomes using the GCM and EGCM. The Figure 6.7 illustrates the correlations between MRI markers and cognitive test scores, as well as their respective relationships with the AD diagnosis. This work aims to reveal additional insights into the progression of AD by leveraging real data from the ADNI (ADNI Team 2023), available through the `Adnimerge` package. Specifically, we examine variables closely

linked to MRI outcomes crucial for AD diagnosis—ventricles X_1 , hippocampus X_2 , and entorhinal cortex X_3 —in conjunction with cognitive test scores: MOCA Z_1 , MMSE Z_2 , and CDRSB Z_3 . These elements are treated as multidimensional variables X and Z , respectively.

Given the early biochemical changes in patients’ brains before a formal AD diagnosis, such as alterations in PET amyloid protein, which serve as a precise standard for confirming AD, we choose the AV45 ratio (Cortical Grey Matter/Whole Cerebellum) as a continuous outcome variable Y (Nordberg et al. 2010; van Maurik et al. 2023). This choice is substantiated by substantial research indicating a significant correlation between the cortical β -amyloid load and early-stage memory decline, thus underscoring the diagnostic importance of AV45 in the early detection of AD (Compta et al. 2012; Bozzali et al. 2011; Yao et al. 2010; Saint-Aubert et al. 2016). The ADNI dataset further illustrates a clear causal relationship between AV45 and AD diagnostic outcomes in Figure 6.7, affirming its utility as an effective proxy for AD diagnosis within this study.

We sidestep this limitation in the GCM framework for categorical data by choosing a continuous variable AV45 ratio instead of a categorical variable AD diagnosis. In the GCM framework, the theoretical foundation relies heavily on the CLT, which is predicated on likelihood functions. However, when the analysis shifts to using log-likelihood, such as with categorical data, the asymptotic conclusions drawn from CLT may not hold. This divergence significantly impacts CI testing, affecting the reliability and accuracy of the results within the GCM framework (Shah & Peters 2020; S. Wood et al. 2017; S. Wood 2017). When dealing with categorical data through log-likelihood, there is a necessity to extend the CLT to log-likelihood case. This extension represents a potential future direction for evolving the GCM framework. In the Section E of the appendix, we will delve deeper into this topic.

This research problem transitions into a statistical CI testing question by aiming to verify whether the structural MRI markers (X) can be considered conditionally independent of the precise standard of AD diagnostic (Y) given the cognitive test scores (Z). In statistical terms, this translates to testing $X \perp\!\!\!\perp Y|Z$, where X represents MRI markers, Y the AV45 ratio, and Z the cognitive test scores. The complexity of this question is heightened by the nature of ADNI data, characterized as repeated measurements, thereby constituting non-i.i.d data. This non-i.i.d aspect poses a significant challenge for many existing CITs, including GCM, which may exhibit bias in the presence of non-i.i.d data. EGCM considers the structure of repeated measurements, positioning it as a potentially effective tool for CI testing within the real-world context of ADNI data. By applying EGCM to ADNI data,

we not only seek to validate its effectiveness in addressing the challenges posed by non-i.i.d data but also to contribute to the broader understanding of AD progression, enhancing the predictive capabilities of MRI markers in the context of existing cognitive assessments.

Our application of GCM and EGCM yields the same results, decisively rejecting the null hypothesis of CI between MRI markers and precious standard of AD diagnostic, considering cognitive test results, with notably low p-values of 0.002. The identical p-values between GCM and EGCM arise from the shared use of the significance threshold $\hat{S}_{n,T}$, defined through the quantile function $\hat{G}$, which facilitates the approximation of $S_{n,T}$'s distribution through Monte Carlo simulation.

The p-value of 0.002 quantifies the evidence against the null hypothesis of CI between the MRI markers and AD diagnostic outcomes, given cognitive test results. Specifically, a p-value of 0.002 indicates strong evidence against the null hypothesis, suggesting that the MRI markers (X) are not conditionally independent of the early AD diagnostics (Y), considering the cognitive assessment tests (Z). In the context of CIT, this result implies a statistically significant association between the structural MRI markers and AD diagnosis outcomes, which cognitive tests alone cannot fully explain. This significant outcome demonstrates that structural MRI data can provide insights beyond those offered by traditional cognitive assessments in predicting early AD diagnostics.

In our study, where multiple structural MRI markers (X) and cognitive test scores (Z) are analyzed to assess their relationship with early AD diagnostics (Y), the issue of multiple testing may arise. This is especially relevant when evaluating the CI of several MRI markers with AD diagnostics, given cognitive scores. Each MRI marker-cognitive score pair essentially constitutes a separate hypothesis being tested for CI from AD diagnostics. As a result, conducting multiple hypothesis tests simultaneously on the same dataset inherently increases the likelihood of encountering Type I errors—false positives, where a test incorrectly rejects the null hypothesis. This inflation of Type I error rate compromises the reliability of statistical inferences made from the tests, potentially leading to misleading conclusions about the relationships between MRI markers, cognitive scores, and AD diagnostics.

Given the multidimensional nature of X and Z , our study was structured to conduct separate CI tests for each dimension of X , explicitly X_1 , X_2 , and X_3 , to mitigate the risk of Type I errors associated with multiple testing. We enable a detailed examination of the individual contributions of each MRI marker to AD diagnostics, given cognitive scores.

This requires the application of False Discovery Rate (FDR) control to adjust the p-values. We use the Benjamini-Hochberg procedure, which aims to limit the expected proportion

of false discoveries (false positives) among all discoveries (rejected null hypotheses). FDR control ensures a more balanced approach to handling Type I error rates across multiple tests in our scenarios. The setup of our study involves segmenting the CI testing into separate testing for each dimension of the structural MRI markers (X) - ventricles ($X1$), hippocampus ($X2$), and entorhinal cortex ($X3$) - given the cognitive test scores (Z). This approach allows us to dissect the intricate relationships between these markers and AD diagnostics, refining our understanding of early AD progression. The study utilizes both GCM and EGCM to perform the CI tests for each MRI marker independently:

- GCM 1 and EGCM 1: $X1 \perp\!\!\!\perp Y \mid Z$,
- GCM 2 and EGCM 2: $X2 \perp\!\!\!\perp Y \mid Z$,
- GCM 3 and EGCM 3: $X3 \perp\!\!\!\perp Y \mid Z$.

This separation into distinct tests for each MRI marker aims to identify which specific MRI markers are most predictive of early AD diagnostics beyond the information provided by cognitive assessments and determine the robustness of the GCM and EGCM in handling multidimensional X and Z in the context of ADNI's repeated measurements and non-i.i.d data structure.

The subsequent analysis and results are synthesized in Table 6.1, presenting the results of GCM tests, while Table 6.2 shows the results of EGCM tests. The application of False Discovery Rate (FDR) control addresses the multiple testing concern, adjusting p-values to account for the aggregate risk of Type I errors. This crucial step ensures the integrity of statistical inferences, particularly when X is multidimensional.

Table 6.1: Results of GCM in Identifying MRI Markers for AD Diagnosis

Test	p -value	Adjusted p -value (FDR control)	Reject
GCM 1	0.0176	1.76×10^{-2}	TRUE
GCM 2	2.02×10^{-9}	6.06×10^{-9}	TRUE
GCM 3	8.37×10^{-4}	1.67×10^{-3}	TRUE

GCM and EGCM both reject the null hypothesis for $X1$ and $X2$, indicating significant correlations between these MRI markers and early AD diagnostics, considering cognitive scores. The adjusted p-values through FDR control maintain the significance of these findings, This provides strong evidence against the null hypothesis for both the ventricles and hippocampus, reinforcing their importance in AD diagnostics beyond cognitive score

Table 6.2: Results of EGCM in Identifying MRI Markers for AD Diagnosis

Test	p -value	Adjusted p -value (FDR control)	Reject
EGCM 1	5.08×10^{-34}	1.52×10^{-33}	TRUE
EGCM 2	6.83×10^{-27}	1.37×10^{-26}	TRUE
EGCM 3	0.899	0.899	FALSE

implications. However, for $X3$ (entorhinal cortex), only GCM indicates a significant relationship, whereas EGCM does not. EGCM 3 presents a stark contrast with a p -value of 0.899, which means not to reject the null hypothesis. This result suggests that, unlike with GCM, EGCM does not find a significant association between the entorhinal cortex volume and AD diagnostics when cognitive tests are considered, highlighting the ability of EGCM to accurately recognize the CI by considering the structure of repeated measurements in the data. These findings coincide with the results of medical studies (Olsen et al. 2017; Scheff & Price 1993), which confirm the direct correlation of changes in entorhinal cortex volumes with cognitive decline impaired by AD.

These results directly address our problem of interest by demonstrating that MRI markers offer valuable insights into early AD diagnostics, potentially enhancing the predictive capabilities of existing cognitive assessments. Specifically, the findings underline the importance of incorporating structural MRI data into the diagnostic process for AD, advocating for a multi-dimensional approach that includes both cognitive and neuroimaging assessments to achieve a more comprehensive understanding of the disease’s progression. The distinction in outcomes for the entorhinal cortex between models also emphasizes the need for advanced methodologies like EGCM to accurately navigate the complexities inherent in ADNI data, thereby contributing significantly to the broader understanding of AD progression.

The examination of residual plots from GCM and EGCM analyses provides critical insights into the dynamics between structural MRI markers, particularly the entorhinal cortex volume ($X3$), and cognitive assessment scores within AD diagnosis. Residual plots are a fundamental diagnostic tool in regression analysis, used to assess the fit of a model to the observed data. In the context of CI testing within AD research, examining residual plots helps identify correlations not captured by the GCM and EGCM. A well-fitted model should produce residuals that are randomly distributed around the zero line, indicating that the model’s predictions are unbiased and consistent across the range of data.

Outliers in residual plots can highlight data points that deviate significantly from the model’s predictions, which could influence the CI testing’s overall performance. In the

GCM, an abundance of outliers can lead to a misinterpretation of the relationship among variables X , Y , and Z . When outliers dominate the residual plots, they could mask the true underlying patterns of dependency between the variables. Specifically, the presence of numerous outliers may skew the residuals' distribution, making it appear as though the residuals are randomly distributed, leading GCM to incorrectly conclude that there is no relationship among X , Y , and Z at all. This situation often results in GCM failing to reject the null hypothesis, which might incorrectly suggest CI among the variables. The residual plot derived from the GCM analysis in Figure 6.8 exhibits noticeable clustering and the presence of outliers, suggesting that the GCM may not fully capture the underlying relationship between the entorhinal cortex volume and AD diagnostics, given cognitive scores. This clustering prompts the rejection of the null hypothesis of CI. In stark contrast, the EGCM analysis shows a more uniformly distributed set of residuals in Figure 6.9, with fewer outliers and no apparent clustering. This pattern indicates a superior fit of the EGCM to the ADNI data, suggesting that EGCM can more accurately account for the complexities inherent in the relationship between X_3 and AD diagnostics, considering cognitive assessments. The absence of correlation in the EGCM's residual plots aligns with the acceptance of the null hypothesis of CI for the entorhinal cortex volume.

The residual plots demonstrate a clear distinction between the GCM and EGCM analyses in terms of handling intra-subject correlations within the ADNI dataset. For the GCM analysis, if the residuals from the same subject exhibit a noticeable correlation, this suggests that the GCM has not fully accounted for the within-subject variability. Such correlations could lead to biased CI testing and invalid inference, as standard GCM assumptions about i.i.d of residuals are violated.

The EGCM is specifically designed to address such non-i.i.d data by incorporating random effects for within-subject correlations. The EGCM residuals plot in Figure 6.9 shows that residuals from the same subject are more independently distributed, this indicates that the EGCM has effectively managed to separate these previously correlated residuals from GCM analysis. The adjustment improves the independence of the residuals, indicating a better model fit that appropriately captures the non-i.i.d. repeated measurements characteristics. This improvement directly impacts CI testing in the context of AD research. The ability of EGCM to handle within-subject correlation means that it can provide a more reliable assessment of the CI between structural MRI markers and AD diagnosis, given cognitive scores.

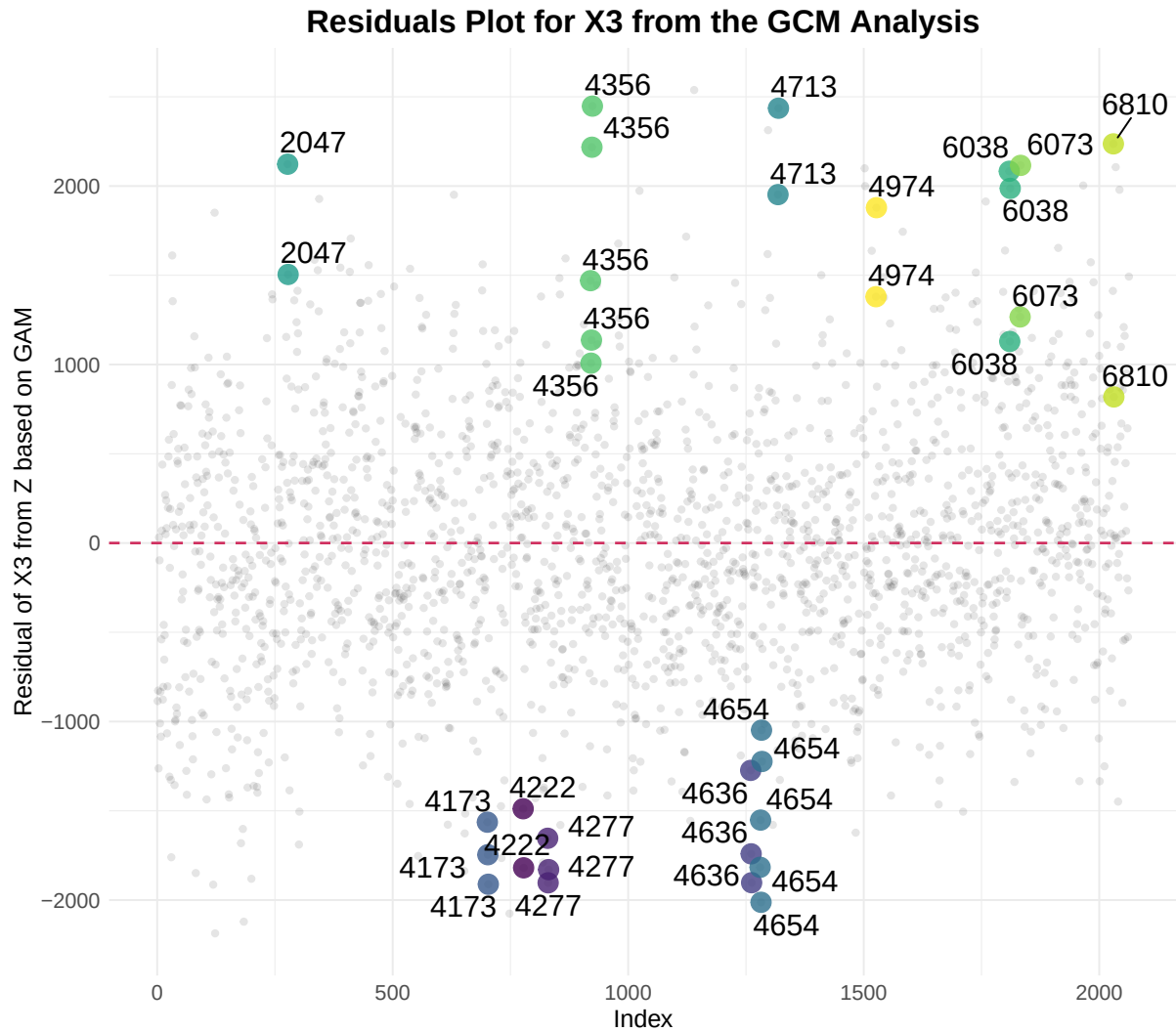


Figure 6.8: This figure presents a residual plot for X_3 derived from GCM. Outliers are accentuated with larger points and annotated with their subject index, whereas the clustering of residuals belonging to the same subjects is indicated by uniform colors. Notably, the clustering of residuals—especially those markedly deviating from the zero line—suggests GCM’s limitations in fully accounting for intra-subject correlation, potentially leading to biased outcomes in CI testing.

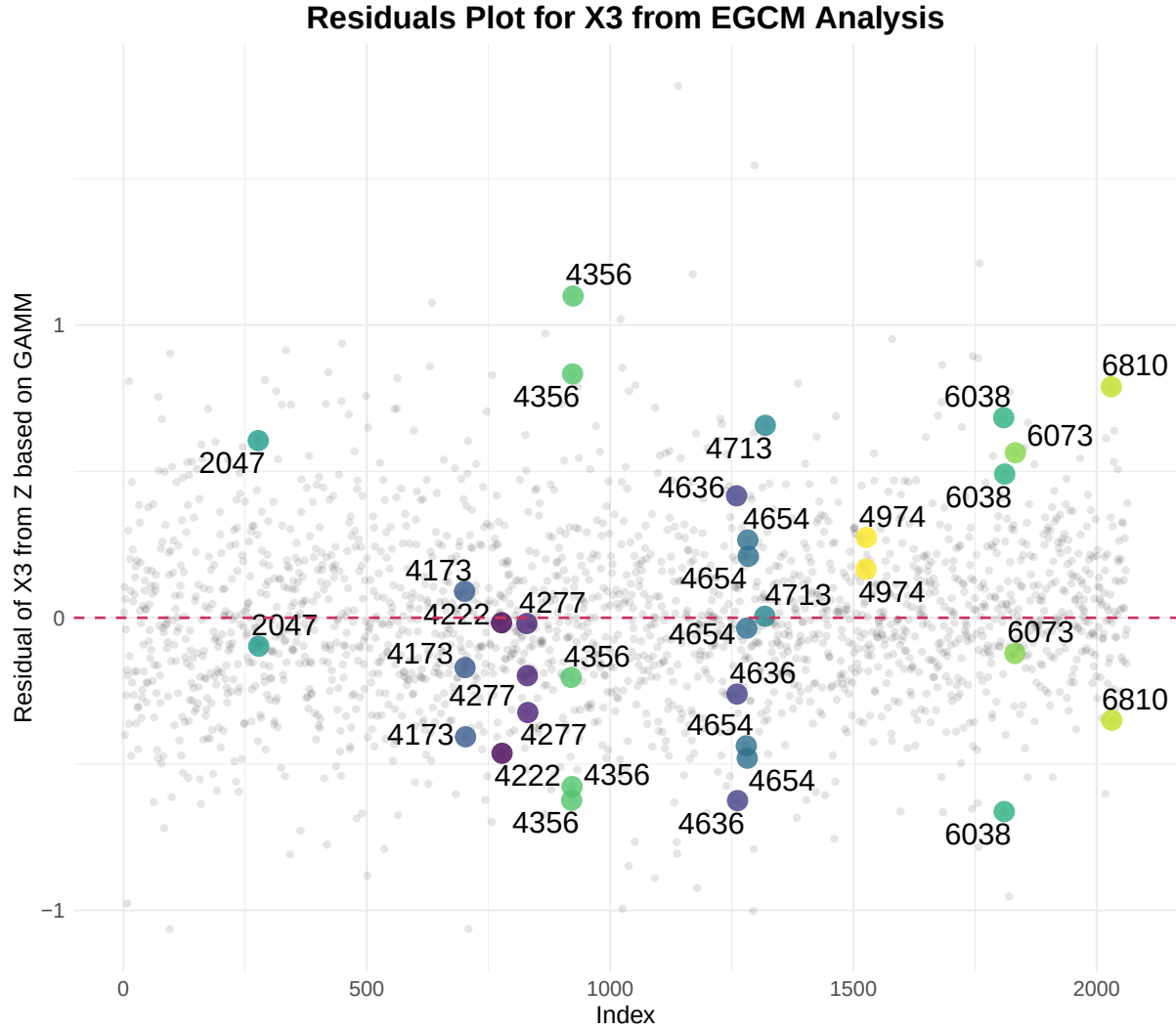


Figure 6.9: This figure depicts the residual plot from the EGCM of X_3 . The residuals' distribution here is markedly more uniform and less dispersed than that observed in the GCM, with outliers being fewer and more proximal to the zero line, indicating an improved model fit. The lack of residual patterns and reduced outlier presence suggest that EGCM achieves a more precise fit for the data, adeptly handling the structure of repeated measurements and enhancing the accuracy of CI testing in AD research.

7 Conclusions and Discussions

In this thesis, we reflect on the challenges encountered in the pursuit of effective CI testing for non-i.i.d data, such as inflated Type I errors and diminished power found in repeated measurements. Our investigation emphasizes the EGCM for its potential to handle the intricacies of repeated measurements data with random effects, which traditional CITs often struggle with.

The research is motivated by the need to improve CI testing methods for non-i.i.d. repeated measurements data, exemplified by ADNI dataset. This is critical for progressing fields like medical diagnostics, where understanding the interdependencies among various biomarkers is key to disease detection and progression tracking. The primary goal is to develop and validate a CI testing approach that can adeptly handle the specific challenges posed by non-i.i.d. data. The EGCM is proposed as a solution to this challenge, aiming to provide more accurate and reliable testing for repeated measurements datasets with random effects.

To address this problem, the thesis explores a range of CITs, including metric-based, kernel-based, simulation-based, and regression-based tests, and demonstrates the diversity of these CITs in capturing conditional dependencies across a broad spectrum of data scenarios. The GCM offers a potential solution for handling non-i.i.d. data, driven by its regression-based test framework that integrates domain-specific regression methods. By skillfully regressing variables against a conditioning set and examining the residuals' covariance to test CI, GCM shows its adaptability across diverse data contexts. An example characterized by repeated measurement with random intercept demonstrates standard GCM can fail in such cases, leading to inflated Type I errors and reduced test power. The EGCM is a significant expansion of the GCM, augmented to account for the random effects present in repeated measurements data. Through an exploration of the possible mixed effects models tailored for non-i.i.d. repeated measurements, we choose GAMMs to extend the GCM for their capacity to model the dependencies and non-linear relationships prevalent in repeated measurements data.

The study engages in comprehensive simulation studies to evaluate the efficacy of the EGCM, comparing it with GCM across various scenarios. The simulations are tailored to

mimic conditions of linear and non-linear relationships, random effects, and even synthetic datasets modeled on real-world data. The results consistently highlight EGCM's superiority in accurately detecting CI, particularly outperforming GCM in repeated measurements with random effects.

In applying the EGCM to actual ADNI data, the thesis successfully translates a complex medical question into a statistical CI testing problem. The EGCM proves effective in uncovering significant relationships between MRI markers and early AD diagnosis that go beyond the explanatory power of cognitive tests. The ability of EGCM to differentiate which MRI markers are meaningful in early AD diagnostics holds significant implications for medical research and practice. It reaffirms the value of incorporating structural MRI data into diagnostic processes and a potential impact on improving the predictive capabilities of existing cognitive assessments.

A key to our analyses has been the thorough examination of residuals. Theoretical considerations underscore the importance of ensuring that residuals, processed through regression models such as GAMM, are i.i.d. with zero mean and finite variance, to maintain the validity of CI testing. The residual analysis allows us to verify the model's assumptions and ensure accurate CI testing. We suggest to integrate a systematic residual analysis step within the GCM and EGCM frameworks. This integration aims to mitigate potential distortions arising from correlated residuals, which might lead to an incorrect interpretation of data interrelationships.

Looking ahead, there are clear pathways for further enhancing the EGCM framework. These include:

- Benchmarking the effectiveness of EGCM against the broader landscape of CITs in terms of Type I error rates, statistical power, and computational efficiency across various simulation settings and real-world datasets.
- Address the limitations of the CLT when dealing with log-likelihood and other complex data transformations, adapting EGCM for efficient handling of categorical data, thus broadening its applicability.
- Integrating more complex GAMMs that account for interactions between random effects, and with state-of-the-art machine learning techniques, such as neural networks within the EGCM framework to enhance its predictive capabilities in high-dimensional or large datasets.

The potential advancements of the EGCM framework proposed in this thesis mark a

7 *Conclusions and Discussions*

significant step towards addressing the unique challenges presented by non-i.i.d. data. By exploiting the potential of EGCM, we pave the way for a more sophisticated and refined approach to CI testing that helps to understand and explore conditional dependencies in non-i.i.d. domains more deeply. The implications for Alzheimer’s research are particularly promising, with the prospect of providing a more detailed diagnostic approach and a better understanding of the progression of the disease.

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Appendix

A. Generalized Covariance Measure Theoretical Framework

The theoretical framework focuses on two primary error metrics, A_f and A_g , and their weighted counterparts, B_f and B_g . These metrics quantify the accuracy of the estimated conditional expectations $\hat{f}$ and $\hat{g}$ compared to the true conditional expectations f_P and g_P , as well as the influence of the conditional variances of the residuals, $u_P(z)$ and $v_P(z)$, on this accuracy. Essentially, they need to diminish as the sample size n increases to maintain the test's validity.

These Mean Squared Prediction Errors (MSPEs) measure the average squared difference between the estimated and actual conditional expectations, quantifying the accuracy of the regression models. The formulas for A_f and A_g are given by:

$$A_f = \frac{1}{n} \sum_{i=1}^n (f_P(z_i) - \hat{f}(z_i))^2, \quad A_g = \frac{1}{n} \sum_{i=1}^n (g_P(z_i) - \hat{g}(z_i))^2,$$

where $f_P(z_i)$ and $g_P(z_i)$ are the true conditional expectations of X and Y given $Z = z_i$, respectively. $\hat{f}(z_i)$ and $\hat{g}(z_i)$ are the estimated conditional expectations from the regression models.

These weighted MSPEs not only assess the predictive accuracy but also factor in the variability of the residuals, providing a more nuanced view of the models' performance. The formulas for B_f and B_g incorporate the conditional variances of the residuals:

$$B_f = \frac{1}{n} \sum_{i=1}^n (f_P(z_i) - \hat{f}(z_i))^2 v_P(z_i), \quad B_g = \frac{1}{n} \sum_{i=1}^n (g_P(z_i) - \hat{g}(z_i))^2 u_P(z_i),$$

where $v_P(z_i)$ and $u_P(z_i)$ represent the conditional variances of the residuals for X and Y , respectively.

The theorem asserts two main conditions under which the test statistic $T^{(n)}$ converges to a standard normal distribution:

- (i) Asymptotic Standard Normality: If the MSPEs and their weighted versions are sufficiently small ($o_P(n^{-1})$ for the product $A_f A_g$ and $o_P(1)$ for both B_f and B_g), and

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if the expectation of the product of squared residuals is finite, then the probability distribution of $T^{(n)}$ closely approximates that of a standard normal distribution as n increases.

$$\sup_{t \in \mathbb{R}} |\mathbb{P}_P(T^{(n)} \leq t) - \Phi(t)| \rightarrow 0,$$

under the conditions that $A_f A_g = o_P(n^{-1})$, and $B_f = o_P(1)$, $B_g = o_P(1)$ are met, indicating the errors diminish appropriately as the sample size increases.

- (ii) **Uniform Convergence:** Under similar conditions for a subset of distributions $\mathcal{P}$ within $\mathcal{P}_0$, with additional constraints on the expectations of squared residuals, the test statistic's distribution uniformly converges to a standard normal distribution across $\mathcal{P}$.

$$\sup_{P \in \mathcal{P}} \sup_{t \in \mathbb{R}} |\mathbb{P}_P(T^{(n)} \leq t) - \Phi(t)| \rightarrow 0,$$

for a subset of distributions satisfying similar diminishing conditions on the errors and certain bounds on the expectations of squared residuals.

The decomposition of the sum $\frac{1}{\sqrt{n}} \sum_{i=1}^n R_i$ into components b , ν_g , and ν_f , plus the sum of the products of residuals $\varepsilon_i \xi_i$, illustrates the influence of prediction errors and the interaction between residuals on the test statistic. This breakdown is vital for understanding the dynamics that underlie $T^{(n)}$'s behavior:

$$\frac{1}{\sqrt{n}} \sum_{i=1}^n R_i = (b + \nu_g + \nu_f) + \frac{1}{\sqrt{n}} \sum_{i=1}^n \varepsilon_i \xi_i,$$

where

$$b := \frac{1}{\sqrt{n}} \sum_{i=1}^n \{f(z_i) - \hat{f}(z_i)\} \{g(z_i) - \hat{g}(z_i)\},$$

$$\nu_g := \frac{1}{\sqrt{n}} \sum_{i=1}^n \varepsilon_i \{g(z_i) - \hat{g}(z_i)\} \quad \nu_f := \frac{1}{\sqrt{n}} \sum_{i=1}^n \xi_i \{f(z_i) - \hat{f}(z_i)\}.$$

b captures the bias introduced by the prediction errors in both f and g . ν_g and ν_f represent the noise components attributable to the interaction between the residuals and prediction errors of g and f , respectively. The term $\frac{1}{\sqrt{n}} \sum_{i=1}^n \varepsilon_i \xi_i$ is central to the Central Limit Theorem (CLT)'s application, indicating that the distribution of the sum of products of residuals (which are presumed to be independent and identically distributed) will tend towards a normal distribution as n grows.

Appendix

The theoretical foundation and proof approach for the Type I error analysis in CI testing using GCM illustrate the critical role of error metrics in assessing the performance of $\hat{f}$ and $\hat{g}$. By ensuring these estimators' accuracy within specified bounds, the framework guarantees the test statistic's convergence to a standard normal distribution, thereby enabling reliable hypothesis testing.

The power of a test is its ability to correctly reject a false null hypothesis. Especially for GCM when there's a nonzero conditional covariance between X and Y given Z . The formula for conditional covariance (Daudin 1980):

$$\mathbb{E}_P(\varepsilon_P \xi_P) = \mathbb{E}_P \text{cov}_P(X, Y|Z)$$

where $\text{cov}_P(X, Y|Z)$ is defined as $\mathbb{E}_P(XY|Z) - \mathbb{E}_P(X|Z)\mathbb{E}_P(Y|Z)$. It mathematically represents the expectation of the product of residuals (ε_P and ξ_P), which are the differences between the observed values and their expected values given Z . It essentially measures the expected deviation of the X and Y products from their conditional means, given Z . Under the null hypothesis H_0 (indicating CI), this value should be zero, implying X and Y do not vary together once Z is accounted for.

A focus on out-of-sample prediction errors A_f and A_g through sample splitting ensures that the estimations for $\hat{f}$ and $\hat{g}$ are not overfitting the data. By developing these estimations from a dataset independent of the primary dataset, the approach ensures that the test's power remains robust against various alternative hypotheses (Shah & Peters 2020). The expected conditional covariance ρ_P and standard deviation σ_P are drfined as:

$$\rho_P = \mathbb{E}_P \text{cov}_P(X, Y|Z), \quad \sigma_P = \sqrt{\text{var}_P(\varepsilon_P \xi_P)}$$

The conditions for the test's power are:

$$\begin{aligned} \sup_{t \in \mathbb{R}} \left| \mathbb{P}_P \left(\frac{\tau_N^{(n)} - \sqrt{n}\rho_P}{\tau_D^{(n)}} \leq t \right) - \Phi(t) \right| &\rightarrow 0, \quad \tau_D^{(n)} - \sigma_P = o_P(1), \\ \sup_{P \in \mathcal{P}} \sup_{t \in \mathbb{R}} \left| \mathbb{P}_P \left(\frac{\tau_N^{(n)} - \sqrt{n}\rho_P}{\tau_D^{(n)}} \leq t \right) - \Phi(t) \right| &\rightarrow 0, \quad \tau_D^{(n)} - \sigma_P = o_P(1). \end{aligned}$$

These conditions involve the convergence of the distribution of the test statistic $\tau_N^{(n)}$, normalized by the difference between $\sqrt{n}\rho_P$ and the denominator $\tau_D^{(n)}$, to a standard normal distribution. These conditions emphasize that for the test to be powerful, the actual variance of the test statistic should closely match the expected variance, and the

estimation of ρ_P should be precise, even as the sample size increases. This precision and convergence are vital for confidently rejecting the null hypothesis when appropriate.

These mathematical principles and testing conditions provide a robust GCM framework for CI testing.

B. Example for Generalized Linear Mixed Models

This section clarifies binomial GLMM example for providing illustrates how these models can be used to link the probability of success in a binary outcome to a set of predictor variables in section 4. The inclusion of random effects allows for the accommodation of individual variability in the response.

GLMMs enable the response variable to be modeled as a smooth, monotonic function of a linear predictor. This approach allows the response to follow a variety of distribution types within the exponential family, such as Poisson, gamma, or binomial. For an illustrative example, consider a binomial GLMM defined by the linear predictor η :

$$Y_{it} \underset{\text{ind}}{\sim} \text{binom} \left(\frac{\exp(\eta_{it})}{1 + \exp(\eta_{it})}, 1 \right)$$

$$\text{where } \eta_{it} = \beta_0 + b_i + \beta_1 x_{1i} + \beta_2 x_{2i}, \text{ and } b_i \underset{\text{ind}}{\sim} N(0, \sigma_b^2).$$

This equation suggests that Y_{it} follows a binomial distribution where the probability of success is linked to the predictor variables via a logistic function $\exp(\eta_i)/\{1 + \exp(\eta_i)\}$. The variability among subjects is introduced through the intercept term b_i . This model's flexibility allows for complex interpretations and the modeling of real-world situations, such as the likelihood of event occurrence in medical trials or consumer behavior studies, where variability across subjects or groups is expected. For example, patient recovery rates in clinical trials, where Y_{it} might represent the success (1) or failure (0) of recovery for the i^{th} patient at time t , and η_{it} includes both treatment effects and patient-specific factors.

In this context, the b_i coefficients, once the variance σ_b is estimated, are referred to as best unbiased linear predictors (BLUPs), conditional modes, or posterior modes (Baayen et al. 2015). This approach estimates individual random effects, offering an intricate insight into subject-specific variations from the overall population mean.

C. Penalized Quasi-Likelihood

The PQL method is a technique designed to manage the inherent complexity of GAMMs by integrating penalty terms into the log-likelihood. This integration aims to regulate the smoothness of the model's functions, preventing overfitting and ensuring that the model remains generalizable to new data. The penalized log-likelihood function is defined as:

$$l_{\text{pen}}(\delta, Q) = \sum_{i=1}^n \log \left(\int f(y_i | \delta, Q) p(b_i, Q) db_i \right) - \frac{1}{2} \lambda \sum_{j=1}^m \alpha_j^T K_j \alpha_j,$$

where λ is a smoothing parameter that controls the degree of penalty applied to the smoothness of the model functions, and K_j are matrices that penalize the smoothness parameters α_j , ensuring the model does not overfit the data. In the case of B-splines, penalties on the differences between adjacent coefficients are applied as $\alpha_j^T K_j \alpha_j = \alpha_j^T (\Delta^d)^T \Delta^d \alpha_j$, with Δ^d being the differential operator matrix of degree d . The covariance matrices for random effects, denoted by Q , may rely on an undetermined parameter vector indicating correlation, with estimates for Q procured through restricted maximum likelihood (REML)-type approximations or other methodologies.

Leveraging the approximation method proposed by Breslow & Clayton (1993), the double penalized log-likelihood becomes:

$$l_{\text{pen}}(\delta, Q) = \sum_{i=1}^n \log (f(y_i | \delta, Q)) - \frac{1}{2} b^T Q_b^{-1} b - \frac{1}{2} \lambda \sum_{j=1}^m \alpha_j^T K_j \alpha_j,$$

with Q_b being a block-diagonal matrix composed of Q . The first penalty term, $b^T Q_b^{-1} b$, arises from the Laplace method approximation, while the second term, $\lambda \sum_{j=1}^m \alpha_j^T K_j \alpha_j$, regulates the smoothness of the $\alpha_j(\cdot)$ functions.

D. Generalized Additive Mixed Model Selection

For the efficacy of the double penalized log-likelihood formula, the selection of penalty parameters necessitates careful consideration. Typically, a procedure such as forward or backward elimination is employed, alongside a criterion like the AIC, BIC, or GCV based on the log-likelihood of the observed data. These criteria facilitate the comparison of various candidate models (H. Wang et al. 2007). However, calculating these criteria becomes challenging in models with random effects due to the dependency on observed data quantities, which may involve complex integrals.

Moreover, H. Wang et al. (2007) showed that even with simple linear models, using GCV might accidentally lead to overfitting. Overfitting in mixed effects models often happens when too many random or fixed effects are included, making the model too complex for the actual data. Rohloff et al. (2022) pointed out that usual model selection criteria might not fully grasp the complexity of models with nonlinear shapes, which could end up in overfitted models. Using techniques like Smoothly Clipped Absolute Deviation (SCAD) (Fan & Li 2001) and Adaptive Least Absolute Shrinkage and Selection Operator (ALASSO) (Zou 2006) helps prevent overfitting. These methods add a penalty for extra parameters, leading to simpler models that are less likely to overfit the data (Ibrahim et al. 2011).

Handling overfitting requires being very careful in choosing models, and paying attention to the complexity of the models' shapes. By carefully using penalty parameters and selection criteria, researchers can achieve a balance, ensuring models are neither overly simplistic nor excessively complex for the given data set.

E. Challenges with Categorical Data

Although GAMMs are theoretically capable of handling categorical data by incorporating smooth functions to capture non-linear relationships between variables, there are significant considerations when extending GAMM to the GCM framework:

The fitting effectiveness of methods based on PQL for categorical data is inherently limited, especially when the data exhibits a high degree of discreteness. This limitation arises because PQL attempts to approximate the data to a normal distribution, a transformation that may not accurately reflect the true structure of discrete data (S. Wood et al. 2017).

Considering the use of GAMM for categorical data within the GCM framework necessitates an expansion of the CLT applicability to log-likelihood situations. This expansion is not only technically challenging but also beyond the scope of current discussions, representing an interesting direction for future research.

The theoretical capability of GAM and GAMM to manage categorical data through logistic regression and other methods underscores the complexity of integrating categorical and continuous data within statistical models. The differences in residuals between categorical and continuous data regressions pose challenges for CI testing, potentially affecting the reliability of outcomes. The nuanced approach required to handle categorical data within GAM and GAMM, as noted in discussions around the `mgcv` and `gamm4` package and its handling of binary and low mean count data, points to the broader challenge of

developing robust statistical models that can accurately capture the complexities of mixed data types (S. Wood 2017; S. Wood & Wood 2015; S. Wood et al. 2017).

Therefore, despite the technical feasibility of GAMMs to process categorical data, the decision to focus on numerical data within the current research context stems from practical considerations regarding the effectiveness of existing methods and the accuracy of statistical inference within the GCM framework. This choice reflects a pragmatic approach to maintaining the precision and reliability of statistical conclusions, leaving the exploration of extending GAMMs to handle categorical data within the GCM framework as an intriguing area for future investigation.

F. Performance under Misspecification

In certain data-generating processes where random effects are particularly strong, as specified below:

$$X_{it} = 3\alpha_i + f(Z_{it}) + (3 + RE_{sx})Z_{it} + 0.3\varepsilon_{it},$$

$$Y_{it} = 3\beta_i + g(Z_{it}) + (3 + RE_{sy})Z_{it} + bf(Z_{it}) + 0.3\xi_{it}.$$

The performance of EGCM is as good as when there is no misspecification in the model. Even under scenarios where only random intercepts were considered (`random = (1|Subject)`), EGCM still showcased a superior performance compared to GCM. Notably, the Type I error rates for EGCM remained significantly lower than those for GCM, indicating a more accurate identification of conditional independence. The results are as follows:

Table A.1: Type I Error Rates under Model Misspecification in GCM and EGCM Simulations

Subject number	GCM	EGCM
50	0.51	0.08
100	0.48	0.12
200	0.49	0.08
300	0.52	0.06
400	0.43	0.08

These tables illustrate that even in the presence of strong random effects within the DGP, and under model misspecifications that fail to account for random slopes, EGCM manages to maintain low Type I errors and high power. This underscores EGCM's robustness and its ability to accurately detect conditional independence across different sample sizes and under varying degrees of model complexity. The results highlight the importance

Appendix

Table A.2: Comparative Power of GCM and EGCM under Model Misspecification in Simulations

Subject number	GCM	EGCM
50	0.52	0.89
100	0.59	0.91
200	0.59	0.88
300	0.45	0.89
400	0.49	0.90

of choosing an appropriate modeling approach, especially in empirical studies where the underlying data structure includes significant random effects.

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Berlin, the 27. March 2024

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